

The sun as colliding beam cosmic ray factory

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Abstract

A theory of cosmic ray production within the solar system is presented. Contrary to existing theories of cosmic ray generation that assume cosmic rays to be extra-galactic in origin, this paper describes cosmic rays as originating primarily within the solar system. *The sun's changing magnetic flux linkage makes it a natural betatron cosmic ray particle acceleration candidate.* Extra-galactic high energy nuclei are, however, essential for initial start-up of this solar cosmic ray mechanism.

The puzzle of how positrons and anti-protons can be well represented even at very high energies, is addressed and accounted for. It is a consequence, initially, of the low energy capture of both particles and anti-particles by the sun's magnetic field, into counter-circulating low energy beams. Later, as the magnet field bending becomes negligible compared to the gravitational bending, both positive and negative beams will have survived the gradual transition from magnetic to gravitational bending. As an aside: this requires a curious "fine-tuning" of the sun's bending fields.

The only role played by the extra-galactic nuclear particles is, when necessary, to "start up" the process from scratch, with the help of the the sun's solar wind. Eventually the self-sustaining atmosphere of cosmic rays currently detectable within the solar system is restored. Perhaps, but not necessarily, this start-up needs to be repeated every 11 years, coinciding with solar magnet and electric field reversals. Certainly the cosmic ray production must have been started up at least once in the past. The cosmic ray energy actually inherited from outside the solar system is negligibly small, compared to the energy subsequently contributed to comic ray generation by the sun itself.

The high quality of cosmic ray data collected over recent decades, at steadily increasing energies, especially by the International Space Station (ISS), make the study of cosmic ray production mechanisms both timely and essential. Highly energetic cosmic ray nuclear particles of all A-values, along anti-protons, and electrons of either sign are observed at present in nature.

The paper proceeds to describe how longitudinal electric fields, explained by the Parker theory of the solar wind, can enable the sun to serve as a "booster" accelerator of cosmic rays, which increases the maximum cosmic ray energies by many orders of magnitude. High energy particle collision processes also maintain the particle abundances at every energy. The dynamic range of energies of the full process produces the observed 13 orders of magnitude maximum particle energy and the energy flux needed to maintain the cosmic ray atmosphere equilibrium.

A steady state mechanism is described, based on semi-quantitative discussion of a relativistic Hamilton-Jacobi formalism, according to which the highest cosmic rays observed can have been produced by the Parker longitudinal electric field component, during fractionally brief, but periodic, semi-circular turns centered on the sun.

1 Introduction

This paper refers to my recent arXiv paper, but already out-dated, due to a significant error concerning the cosmic ray start-up mechanism. It is available in the arXiv [1]; a paper that emphasizes laboratory-based astrophysical experiments while this paper discusses only solar cosmic ray production. To access the previous arXiv paper, click on the link: "Appendix"

The present paper is based on recent experimental cosmic ray studies, mainly aboard the International Space Station (ISS). Some figures specific to cosmic ray production are common, often modified, to both papers. There are no lengthy tables in the present paper. In this respect the arXiv paper can be regarded as an "appendix" to the present paper.¹

¹"High energy accelerator physics units" are employed in the present paper. as in $v = \beta c$. But the units are "natural" in another sense; namely because the value of c can be taken to be "1.0" in numerical evaluations, should the reader's preference be the same as the author's. Planck's (modified) constant " $\hbar$ " does not appear, but if it did, its numerical value would also be "1.0". These conventions cause magnetic units to be confusing, causing them to be avoided to the extent possible. The numerical values of energies will normally be expressed in GeV units. "Joules" will (almost) never appear, and "ergs" never. Any errors or actual inconsistency resulting from this convention should be flagged as typographical errors. Units, especially magnetic, are discussed in greater detail in Appendix A.

Another significant difference is that this paper emphasizes protons, $Z=1$, $N=0$, while that paper emphasizes “alpha-particle-like” nuclear isotopes with equal numbers of protons and neutrons, $Z=N$, $A=2Z$; in particular, *not protons*. A reason for the $Z=N$ “alpha-like” emphasis in the arXiv paper is that all such nuclei have identical “magnetic rigidities”, meaning they have identical orbits in magnetic fields. This is helpful for understanding the Parker solar wind theory, but is probably unimportant in present context. In this paper it is the solar wind (serving as source of particles for acceleration) that is emphasized.

In this paper, once high energy proton cosmic rays are understood, so also will be all nuclear isotope cosmic rays particles, as well as electrons and positrons. Figure (1) shows a range of three topologically similar orbit shapes of nuclear particles being pulled gravitationally toward the sun. The important case is on the left. Only two full cork-screw rotations more or less in

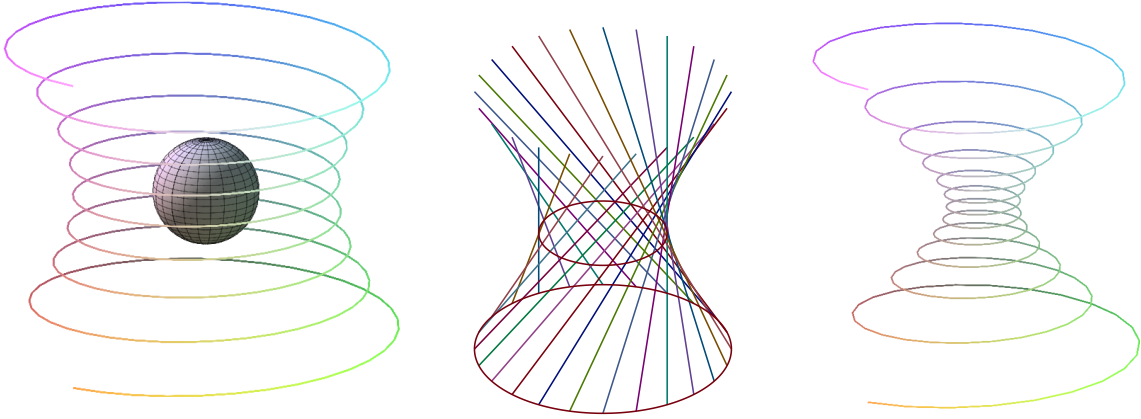


Figure 1: Selected topologies of particles falling toward the sun: **Left:** corkscrew-shaped orbit, “captured briefly” but then released; **Center:** grazing the sun; **Right:** falling into the sun.

the equatorial plane of the sun are shown, but there could be many more. This is the solar cosmic ray geometry to be described and analyzed in this paper.

The Parker model of solar wind

In 1926, A. A. Milne, a British writer, famously penned a puzzle in the form of a verse; “No one can tell me, Nobody knows, Where the wind comes from, Where the wind goes.” Milne was referring, presumably, to the meteorological earth wind. Though a mathematician by training, Milne had not researched this very carefully, since, at that date, the earth’s wind was fairly well understood. In 2025 the Milne puzzle can be applied to the solar wind. As to “where the wind comes from” Milne has again become outdated; Parker has explained this. But “Where the solar wind goes” remains an active puzzle for astrophysics.

The challenge for this paper is to demonstrate that the solar wind can be treated as the injected beam, into an accelerator consisting of the the sun itself, or from a chain elements of the solar system, to produces ultra-high energy cosmic ray particles, all within the solar system.

Solar magnetic fields

Magnetic fields in the interior of the sun are hopelessly chaotic. Interior magnetic field structures, referred to as “flux tubes” are distributed throughout the sun’s interior. These are in the form of more or less cylindrical, randomly oriented, structures that are quite well modeled in isolation. Fortunately, for present purposes, for reasons to be explained, it will not be necessary to understand how the internal stellar matter is organized. Just the magnetic fields at the surface of the sun are sufficient, as boundary conditions, to establish the external magnetic fields. It is, of coarse, important for these surface fields to be reliably known.

Magnetic fields external to the sun are shown in Figure 2. Since magnetic flux lines are continuous the average magnetic fields, even in the interior of the sun, would seem to be required

to match the magnetic fields shown in the upper two plots in Figure 2. It is clear, in the bottom picture, that the surface itself is represented by actual data, made available from the NASA's Goddard Space Flight Center. From their description it seems likely that the external magnetic field lines have been constrained to best fit the available measurements to the theoretically well known magnetic dipole pattern outside the sun. The field outside a uniformly magnetized sphere is the same as the field of an equivalent point dipole located at its center;

$$B(r, z) \Big|_{z=0} = \frac{\mu_0}{4\pi} \frac{(AI)_{\text{sun}}}{(z^2 + r^2)^{3/2}} \Big|_{z=0}; \text{ for } r > R_{\text{sun}}, \quad (1)$$

where $(AI)_{\text{sun}} = 2\pi R_{\text{sun}}^2 I_{\text{eff.}}$, and $I_{\text{eff.}}$ is the effective current flowing along the sun's equator that would produce the observed magnetic field in the free space outside the sun. This is the magnetic field that affects the acceleration of particles during their temporary storage and acceleration while passing by the sun, possibly for multiple turns, until their inevitable nearly isotropic two dimensional escape into a horizontal plane.

These magnetic fields need to be treated carefully. Their role in the Parker solar wind theory, that explains the source of longitudinal electric field which accelerates nuclear isotopes, is essential. At the same time it must be confirmed that any radial magnetic force, either centripetal or centrifugal, does not compete too seriously with the centripetal gravitational pull on (charged) nuclear isotopes circulating above the sun's equator. This region is illustrated and labeled in Figure 2. When referring to the sun as a storage ring or accelerator, the "aperture" is shown in the figure. Of course, no vacuum chamber is required, and the "acceptance" is huge.

Solar electric fields

The top image in Figure 3 is copied from Owens and Forsythe[2]. The bottom figure breaks out the electric and magnetic field patterns. The so-called "Parker spiral angle", θ_s , is indicated by all the arrows in the bottom figure. (The fact that these angles seem not quite equal is a defect of the figure, not of the theory.) The original version of this figure is contained (as a simple sketch) in Parker's original paper explaining the solar wind.

The tangential electric fields result from the 22 year periodic variation of the sun's magnetic polarity. The tangential electric field causes the sun to act as a betatron accelerator. The electromotive force resulting from Faraday's law produces the tangential electric fields shown in the lower figure. The original caption to the figure referred to the figure as

"a sketch of the steady-state solar magnetic field in the ecliptic plane. Close to the sun, in a spatial region approximately bounding the solar corona, the magnetic field dominates the plasma flow and undergoes significant non-radial (or super-radial) expansion with height. At the source surface, typically taken to be a few solar-radii, the pressure-driven expansion of the solar wind dominates and both the field and flow become purely radial. In the heliosphere, rotation of the heliospheric magnetic field (HMF) footprints within a radial solar wind flow generates an azimuthal component of the HMF, B_ϕ , leading to a spiral geometry. Regions of opposite HMF polarity, shown as red and blue, lines, are separated by the heliospheric current sheet (HCS), shown as the green dashed line. Image adopted originally from Schatten, Wilcox, and Ness."

As complicated as it is, Figure 3 requires some qualification in order to be comprehensible. Though a nuclear isotope orbit is sometimes said to coincide with a magnetic field line, *this is categorically incorrect*. The isotope orbits are actually helical, winding around a "guiding center", just as in the earth's Van Allen belts. It is the guiding center that coincides with a magnetic field line.

Ever since the first reported observation of cosmic rays (by Victor Hess in 1912) their source has never been unambiguously identified. A well known first attempt by Fermi, has only been grudgingly accepted, even by Fermi himself. But Fermi's mechanism has commonly favored the view that cosmic ray production is a galactic or extra-galactic phenomenon.

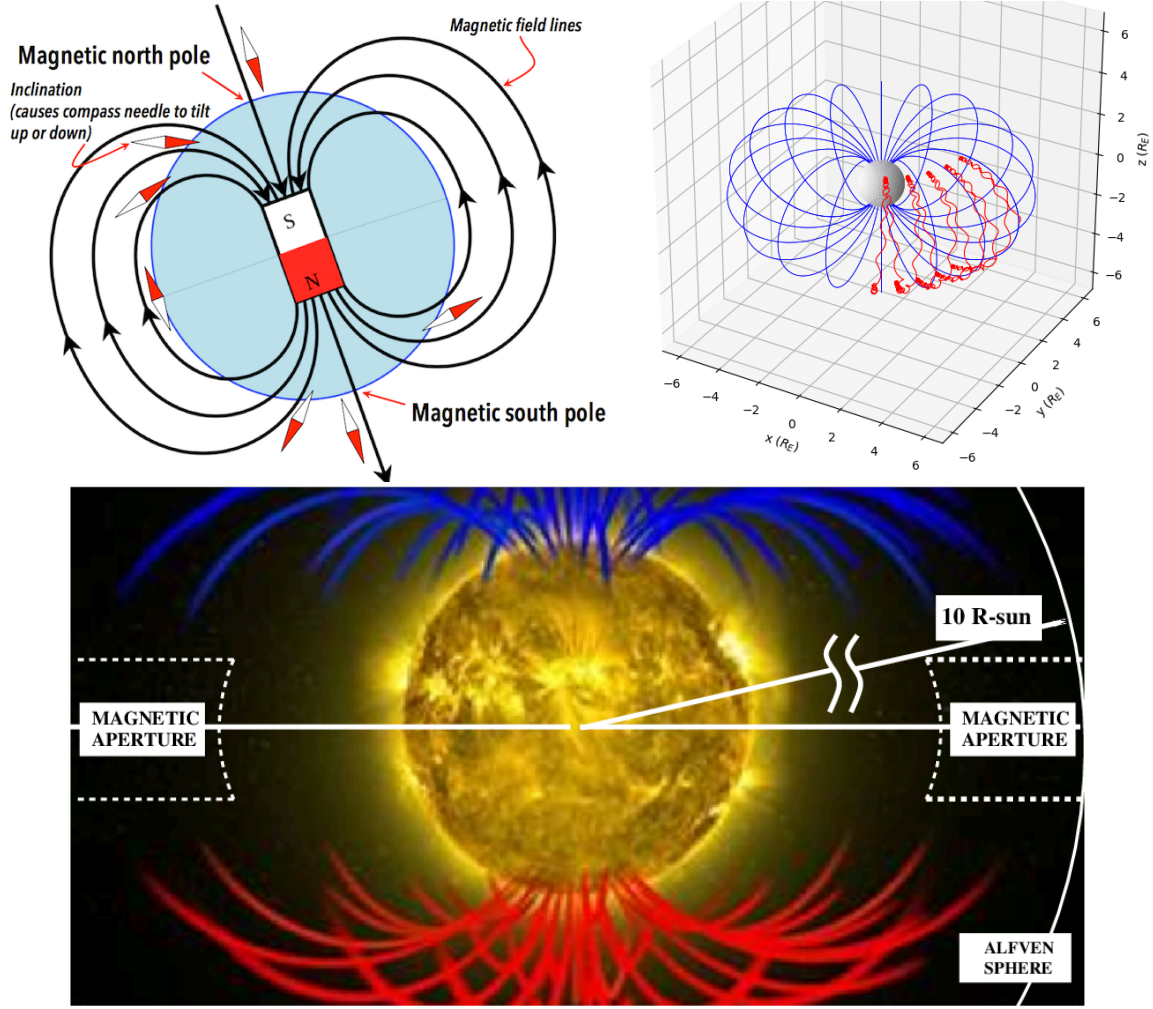


Figure 2: **Top left:** Magnetic dipole field pattern. **Top right:** Perspective view of dipole field pattern. 16 field lines are shown (actually +1, including the straight line from observer's view point). **Bottom:** 2025 image shows magnetic fields radiating from the sun's poles. Courtesy of NASA's Goddard Space Flight Center. Superimposed is the outline of the aperture of the sun as a particle accelerator. Interpolated onto the photograph are outlines of a virtual vacuum chamber for the sun as particle accelerator. Also indicated, at approximately 15 times the sun's radius, is the Alfvén radius, which figures prominently in the sun's injection, acceleration, and extraction processes.

The sun as a betatron

The first fully successful circular particle accelerator was the “betatron” invented by Doald Kerst, at the University of Chicago, just before the Second World War. The source of the particle acceleration was the Faraday's law electromotive force (EMF) induced by the time-varying magnetic field linked by the toroidal vacuum chamber of the ring.

There is a very substantial changing magnetic flux associated with the 22 year periodic variation of the sun's magnetic dipole moment (MDM). The sun's changing magnetic flux linkage makes it a natural betatron cosmic ray particle accelerator.

Faraday's law determination of the sun's maximum betatron EMF

We wish to find the Faraday EMF encountered in a nuclear trip around the sun. The Sun's time-dependent, 22 year periodic magnetic dipole moment variation is the source of time variation.

$$\mu_{\text{sun}}(t) \approx 3.9 \times 10^{33} \text{ Am}^2 \cos(\omega_{\text{sun}} t), \quad (2)$$

HELIOSPHERIC MAGNETIC FIELD (HMF)

EFFECTIVE RADIAL MAG. FIELD SOURCE

Heliosphere
 $\vec{B} = B_R \hat{R} + B_\phi \hat{\phi}$
 $\vec{V} = V_R \hat{R}$

Source surface
 $\vec{B} = B_R \hat{R}$
 $\vec{V} = V_R \hat{R}$

Super-radial expansion
 $\vec{B} = B_R \hat{R} + B_\theta \hat{\theta} + B_\phi \hat{\phi}$
 $\vec{V} = V_R \hat{R} + V_\theta \hat{\theta} + V_\phi \hat{\phi}$

HELIOCENTRIC CURRENT SHEET MAGNETIC FIELD REVERSAL

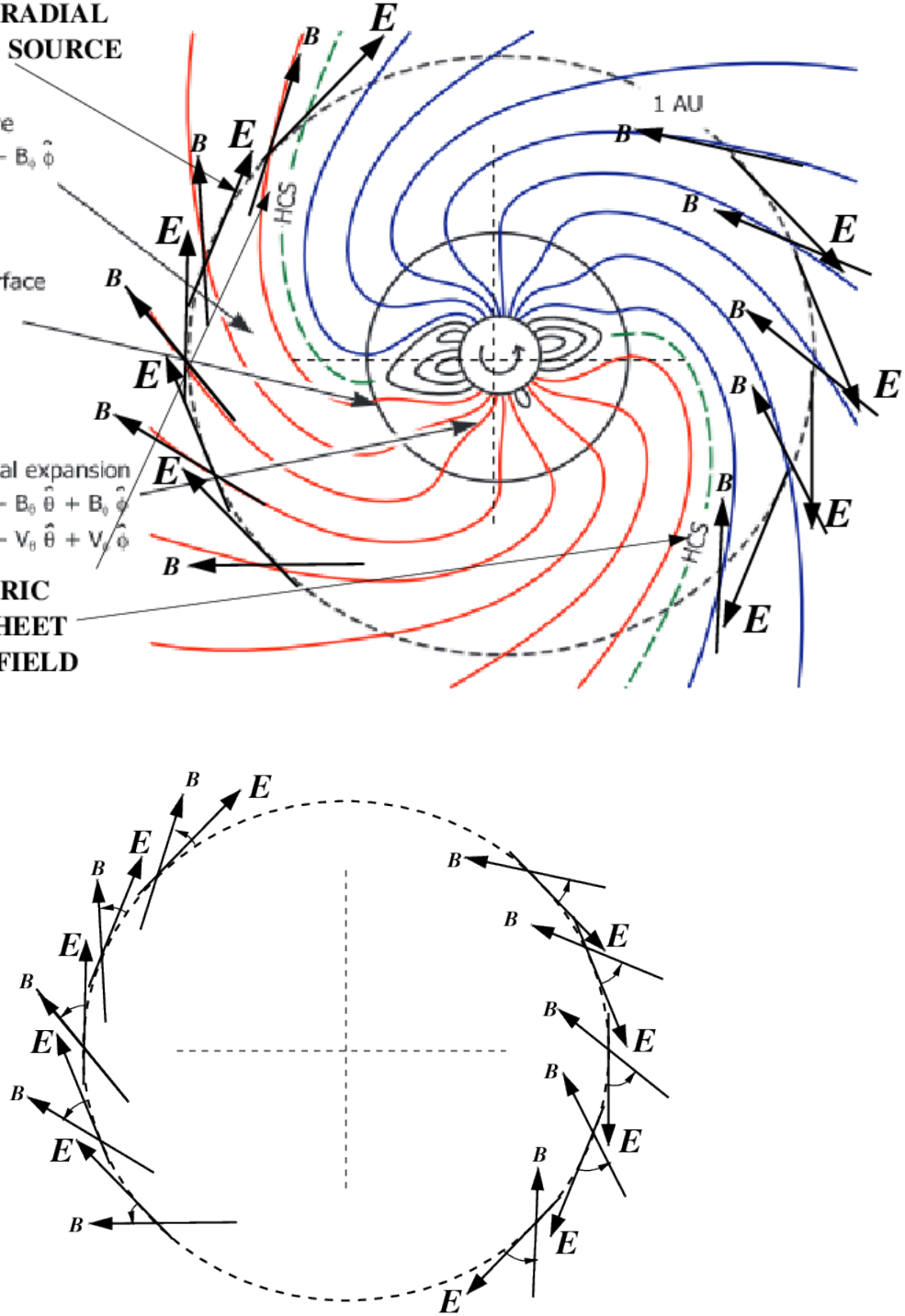


Figure 3: **Top:** Copied and annotated figure from Owens and Forsythe[2], showing magnetic field directions, along with added electric field directions. The electric field directions correspond to the Faraday's law electromotive force resulting from the 22 year period, time-varying axial solar magnetic flux. **Bottom:** Copied from the top figure, the local electric and magnetic field directions are shown, illustrating, for example, the magnetic field reversal across the heliocentric current sheet (HCS). This figure defines the Parker angle θ_P orienting the B -field lines relative to the effective radial magnetic field source and demonstrates its constancy."

where $\omega_{\text{sun}} = 22 \times 10^7 \times 0.31536 \text{ s}$. The maximum time rate of change of μ_{sun} is

$$\frac{d\mu_{\text{sun}}}{dt} = \frac{3.9 \times 10^{33}}{22 \times 10^7 \times 0.31536} \text{ Am}^2/\text{s}. \quad (3)$$

The area of the equatorial plane of the sun is $A_{\text{sun}} = \pi R_{\text{sun}}^2 = 3.14 \times 0.6957^2 \times 10^{18}$. Another formula for the sun's magnetic dipole moment is $\mu_{\text{sun}} = \mu_0 I_{\text{loop}} A_{\text{sun}}$, or

$$\frac{dI_{\text{loop}}}{dt} = \frac{1/\mu_0}{3.14 \times 0.6957^2 \times 10^{18}} = \frac{3.9 \times 10^{33}}{22 \times 10^7 \times 0.31536} \text{ A/s} \quad (4)$$

where $\mu_0 = 4\pi \times 10^{-7} \text{ H/m}$ (Henries/meter a.k.a. T m/A) where I_{loop} is the current in a loop around the equatorial diameter having the same MDM.

For a thin ring of radius that of the sun let us take the wire radius to be $r=1 \text{ m}$, in which case the inductance is given by

$$L = \mu_0 R \ln \left(\frac{8R}{1} - 1.75 \right) = 4\pi \times 10^{-7} \times 0.6957 \times 10^9 \ln(0.55656 \times 10^{10}) = 4 \times 3.159 \times 69.57 \times 22.44 = 2 \times 10^4 \text{ H} \quad (5)$$

The maximum instantaneous Faraday law EMF is the rate of change of the magnetic flux through the inductor using the formula,

$$\mathcal{E} = -2 \times 10^4 \times \frac{dI_{\text{loop}}}{dt} = -2 \times 10^4 \times \frac{3.9 \times 10^{33}}{22 \times 10^7 \times 0.31536} \text{ A/s} \quad (6)$$

Disambiguation of the term “cosmic ray”

Protons are essentially the only hadronic particles escaping from the solar system. Furthermore, these escaping protons can be conjectured to be being replaced (on the average) by immigrating protons from other stars. (To the very limited extent that other light nuclear particles, such as deuterons, escape, they can be treated in the same way as escaping protons.) On a hierarchy of solar, then galactic, then cosmic, sequentially increasing distance and time scales, there can be balanced immigration and emigration of protons, on solar, then galactic, then cosmic time scales.

As regards cosmic rays, my original strict policy for this paper was to discuss only cosmic ray acceleration within the solar system, treated as a closed and isolated system. But, to produce a “start-up” mechanism, I have taken advantage of a modest flux of multi-GeV, extra-galactic origin, confirmed with reliable observations, in supernova explosions.

An unfortunate consequence of my policy is that it renders the term “cosmic ray” *hopelessly ambiguous*. Ever since their discovery in 1912 by Victor Hess, it has been mainly assumed that all particles “coming out of the sky” originated outside the solar system. This is very nearly opposite to the policy in this paper, which treats all “cosmic-rays” as a solar phenomenon.

Though confusing, this is not unprecedented. It simply means that one must admit the possible existence of two classes of particles coming out of the sky. One is “solar system cosmic rays”, the other “non-solar system cosmic rays”. It seems certain that both classes will consist primarily, of protons, deuterons, and α -particles, in that order.

Regrettably, since cosmic ray identifications never establish their source, it must always be accepted that what has been measured is an incoherent sum of solar and non-solar cosmic rays.

It is my prejudiced belief that most “cosmic rays” are “solar”.

As an aside, it can be mentioned that golf ball sized ice balls routinely come out of the sky during electrical storms in Texas. Hundreds of golf ball size pockmarked automobiles guarantee that this is not “fake news”. Apparently this phenomenon is caused, in the presence of electrical storms, by vertical winds, that temporarily prevent the ice balls from falling. In the presence of water vapor, and extremely low temperature, it is no surprise that the ice balls grow quickly.

One supposes that a similar phenomenon can occur when the solar wind holds up counter-traveling nuclear ions long enough for partially charged atomic balls containing radioactive nuclear isotopes to join. The real mystery is how such balls can subsequently retain or recover semi-relativistic velocities high enough to produce μ -mesons, or other incontrovertible evidence of their nuclear history.

2 Astronomical bodies as stochastic accelerators

Rough parameter values

During isolated collisions between two nuclear particles the “strong” nuclear force competes with the “medium strength” electric force. At nuclear scale the nuclear force wins; at atomic scale the electric force wins. At astronomical scale, because of charge neutrality (cancellation of nuclear and electron charge densities to a first approximation) the competition shifts to gravitation versus magnetism.

As earth observers we are poorly equipped to assess any of these forces. With magnetic compass we can detect, in round numbers, a half Gauss earth magnetic field, which is amply strong enough for a light needle on a decent bearing to measure the field direction—in other words negligibly small for doing real work. As it happens, a typical magnetic field at the surface of the sun is not very different—only several Gauss, with the further complication of varying more or less sinusoidally with a period of 22 years.

But, by no means, may the sun’s magnetic field be ignored, since the sun’s radius exceeds the earth’s radius by five orders of magnitude.

“Storage rings” such as the Parker ring around the sun, or the earth’s Van Allen belts, store particles with momentum proportional to the ring radius. With the sun/earth radius ratio being 10^5 , and the momentum of a 2 MeV electron in the Van Allen belt around the earth being possible, the momentum of an electron or proton stored in the Parker ring around the sun could be 200 GeV.² As a matter of fact, the essence of this paper is that protons (and other nuclear isotopes) can acquire energy greater than 200 GeV, in the solar system See Figure 11, copied from chapter 29 of the Particle Data Group report[3].

Jupiter could also serve as a curiously powerful particle accelerator. Its surface magnetic field is nearly the same as the sun, and its radius almost exactly 1/10 as great. Though the radial dependence follows an inverse cube ($1/r^3$) law for the magnetic field strength, it is effectively constant (weakly defocusing) for elevations small compared to Jupiter’s radius. With injection from the solar wind, in the equatorial direction and bending by the centripetal gravitational force, the beam of charged particles circulate in a belt-shaped orbit directly above the equator.

In traditional accelerator coordinates, if the x (horizontal) axis is chosen to point toward the center of Jupiter, and the y (vertical) coordinate points north, then the z (longitudinal) coordinate points east. The gravitational force then points along the positive x axis, and a beam of positive charges travels in the positive z direction. Then the $\mathbf{v} \times \mathbf{B}$ cross product of velocity and magnetic field points toward the center of Jupiter. In this case, the gravitational and Lorentz magnetic forces on a (necessarily positive) nuclear isotope can be said to be “constructive” or, otherwise “destructive”.

Along with the latitudinal electric field provided by the Parker mechanism, Jupiter could serve as a suitable injector for a cosmic particle accelerator complex. Currently, the highest energy terrestrial accelerator (the LHC at CERN) has a maximum proton energy of 7000 GeV, with an average magnetic field of about 6 Tesla.

Somewhat similar considerations apply to the relative importance of gravitational and magnetic forces in stars. Cosmologically natural magnetic fields, either on the earth’s surface or on the surface of the sun are conveniently quoted in Gauss units.

Circular, gravitational, proton orbits around the sun

The major subject of this paper concerns orbits of nuclear particles around the sun for which the bending is due, primarily, to the sun’s gravitational attraction. For the occasional “sanity check” of derived formulas it is useful to establish a trusted calculation to be referred back to

²In this calculation an electron has been chosen to avoid needing to use an exact relativistic formula for the deuteron in an only semi-relativistic context. Both particle energies are large enough for their momenta to be roughly proportional to their energy. In both cases referring to spherical bodies as accelerators is misleading. Actual accelerators have apertures of, perhaps 10 cm. Astrophysical accelerators require no beam tubes, and have nearly unlimited (one-sided) apertures.

occasionally. The (presumably non-relativistic) circular proton orbit around the sun, along the sun's equator, calculated using MKS units, provides such a check. The value for the gravitational acceleration on earth is $g_{\text{earth}} = 9.8 \text{ m/s}^2$. The needed quantities for the sun are $R_{\text{sun}} = 6.963 \times 10^8 \text{ m}$, $g_{\text{sun}} = 273.8 \text{ m/s}^2$, $m_{\text{sun}} = 1.989 \times 10^{30} \text{ kg}$, and proton mass, $m_p = 1.672\,621\,925\,95 \times 10^{-27} \text{ kg}$, all of which should be reconfirmed, before being applied.

According to Newton's various laws one has, for the centripetal force, applied to a proton traveling on an orbit along the equator of the sun is

$$F_p = m_p g_{\text{sun}} = m_p \frac{v_p^2}{R_{\text{sun}}}, \quad \text{or} \quad v_p^2 = g_{\text{sun}} R_{\text{sun}}. \quad (7)$$

The proton velocity and the proton kinetic energy, KE , are then given by

$$\begin{aligned} v_p &= \sqrt{273.8 \times 6.963 \times 10^8} = 4.3663 \times 10^5 \text{ m/s}, \quad (\text{or } \beta_p \approx 10^{-3}), \\ KE_p &= m_p v_p^2 / 2 = 1.67262 \times 10^{-27} \times 273.8 \times 6.963 \times 10^8 / 2 = 1.5994 \times 10^{-15} \text{ J}. \end{aligned} \quad (8)$$

To convert from Joules to GeV one multiplies by 6.242×10^9 , with the result

$$KE_p = 1.5994 \times 10^{-15} \times 6.242 \times 10^9 = 9.984 \times 10^{-6} \text{ GeV or } 10.0 \text{ keV}, \quad (9)$$

which confirms our expectation that the result would be non-relativistic. Furthermore, it provides an easy to remember mnemonic. To be categorized as a "healthy" cosmic ray (meaning "effectively fully relativistic"), such protons would need to be accelerated by six orders of magnitude, to 10 GeV, for example.

One can also refresh one's memory by calculating the gravitational potential on the sun's equator, using Newton's gravitational formula $V_G = -\frac{GM}{R}$.

$$\begin{aligned} V_G \Big|_{\text{sun-surface}} &= -\frac{GM_{\text{sun}}}{R_{\text{sun}}} \\ &= -\frac{6.674 \times 10^{-11} \text{ m}^3/\text{kg} \cdot \text{s}^2 \times 1.989 \times 10^{30} \text{ kg}}{6.963 \times 10^8 \text{ m}} \\ &= -1.906 \times 10^{11} \text{ m}^2 \text{ s}^{-2} \end{aligned} \quad (10)$$

$$\equiv -1.906 \times 10^{11} \text{ J/kg}. \quad (11)$$

Superimposed E,M, or G,M, circular bending formalism

In a storage ring with superimposed E&M bending, the circulation direction of a conventionally-named "master beam" (of whatever charge q_1) is assumed to be CW or, equivalently, momentum $p_1 > 0$. A "secondary" beam charge q_2 is allowed to have either sign, and either CW or CCW circulation direction.

A design particle has mass $m > 0$ and charge qe , with electron charge $e > 0$ and $q = \pm 1$ (or some other integer). These values produce circular motion with radius $r_0 > 0$, and velocity $\mathbf{v} = v\hat{\mathbf{z}}$, where the motion is CW (clockwise) for $v > 0$ or CCW for $v < 0$. With $0 < \theta < 2\pi$ being the cylindrical particle position coordinate around the ring, the angular velocity is $d\theta/dt = v/r_0$.

(In MKS units) qeE_0 and $qe\beta cB_0$ are commensurate forces, with the magnetic force relatively weakened by a factor $\beta = v/c$ because the magnetic Lorentz force is $q\mathbf{v} \times \mathbf{B}$. By convention e is the absolute value of the electron charge; where it appears explicitly, usually as a denominator factor, its purpose in MKS formulas is to allow energy factors to be evaluated as electron volts (eV) in formulas for which the MKS unit of energy is the joule. Newton's formula for radius r_0 circular motion, expressed in terms of momentum and velocity (rather than just velocity, in order to be relativistically valid) can be expressed using the total force per unit charge in the form

$$\beta pc/e = (E_0 + c\beta B_0) qr_0, \quad (12)$$

Coming from the cross-product Lorentz magnetic force, the factor $q\beta cB_0$ is negative for backward-traveling orbits because the β factor is negative.

A “master” or primary beam travels in the “forward”, CW direction. For the secondary beam, the β factor can have either sign. For $q = 1$ and $E_0 = 0$, formula (12) reduces to a standard accelerator physics “cB-rho=pc/e” formula. For $E_0 \neq 0$ the formula incorporates the relative “bending effectiveness” of E_0/β compared to cB_0 . As well as fixing the bend radius r_0 , this fixes the magnitudes of the electric and magnetic bend field values E_0 and B_0 . To begin, we assume the parameters of a frozen spin “master”, charge qe , particle beam have already been established, including the signs of the electric and magnetic fields consistent with $\beta_1 > 0$ and $p_1 > 0$. In general, beams can be traveling either CW or CCW. For a CCW beam both p and β have reversed signs, with the effect that the electric force is unchanged, but the magnetic force is reversed. The β velocity factor can be expressed as

$$\beta = \frac{pc/e}{\sqrt{(pc/e)^2 + (mc^2/e)^2}}. \quad (13)$$

Eq. (12) becomes

$$\frac{pc}{e} = \left(\frac{E_0 \sqrt{(pc/e)^2 + (mc^2/e)^2}}{pc/e} + cB_0 \right) qr_0. \quad (14)$$

Cross-multiplying the denominator factor produces

$$\left(\frac{pc}{e} \right)^2 = qE_0 r_0 \sqrt{(pc/e)^2 + (mc^2/e)^2} + qcB_0 r_0 \frac{pc}{e}. \quad (15)$$

To simplify the formulas we make some replacements and alterations, starting with

$$pc/e \rightarrow p, \quad \text{and} \quad mc^2/e \rightarrow m, \quad (16)$$

The mass parameter m will be replaced later by, m_p , m_d , m_{tritium} , m_e , etc., as appropriate for the particular particle types, proton, deuteron, triton, electron, helion, etc.. These changes amount to setting $c = 1$ and switching the energy units from joules to electron volts. The number of ring and beam parameters can be reduced by forming the combinations ³

$$\mathfrak{E} = qE_0 r_0, \quad \text{and} \quad \mathfrak{B} = qcB_0 r_0. \quad (17)$$

After these changes, the closed orbit equation has become

$$p_m^4 - 2\mathfrak{B}p_m^3 + (\mathfrak{B}^2 - \mathfrak{E}^2)p_m^2 - \mathfrak{E}^2 m^2 = 0, \quad (18)$$

an equation to be solved for either CW and CCW orbits. The absence of a term linear in p_m suggests the restoration, using Eq. (17), of the explicit form of $\mathfrak{B}$ in the coefficient of the p_m^3 term to produce;

$$p_m^4 - 2cB_0(qr_0)p_m^3 + (\mathfrak{B}^2 - \mathfrak{E}^2)p_m^2 - \mathfrak{E}^2 m^2 = 0, \quad (19)$$

The product factor (qr_0) can be altered arbitrarily without influencing any essential conclusions. This and other properties can be confirmed by pure reasoning, based on the structure of the equation, or by explicit partially-numerical factorization of the left hand side.

These considerations have removed some, but not all of the sign ambiguities introduced by the quadratic substitutions used in the derivation of Eq. (19). The electric field can still be reversed without altering the set of solutions of the equation. Note that this change cannot be compensated by switching the sign of q , which also reverses the magnetic bending. The most significant experimental implication is that it is not only positrons, but also electrons, that can have orbits identical to (usually positive in ordinary practice) baryons.

We can contemplate allowing the signs of E_0 or B_0 to be reversed for observational purposes, such as interchanging CW and CCW beams, or replacing positrons by electrons.

³Yet another font, for $\mathfrak{E}$ and $\mathfrak{B}$, has been introduced to represent the scaled values of squares of electric and magnetic fields, without clashing with energies or electric or magnetic fields. It is worth noticing that Eq. (15) depends on B_0 and E_0^2 and B_0^2 , but *not on* E_0 un-squared. This influences the variety of solutions to the quartic equation.

Fractional bending coefficients η_E and η_m can be defined by

$$\eta_E = \frac{qr_0}{pc/e} \frac{E_0}{\beta}, \quad \eta_M = \frac{qr_0}{pc/e} cB_0, \quad (20)$$

neither of which is necessarily positive. These fractional bending fractions satisfy

$$\eta_E + \eta_M = 1 \quad \text{and} \quad \frac{\eta_E}{\eta_M} = \frac{E_0/\beta}{cB_0}. \quad (21)$$

The “potency’s” of magnetic and electric bending are in the ratio $cB_0/(E_0/\beta)$ because the electric field is stronger than the magnetic by the factor $1/\beta$ as regards bending charge q onto an orbit with the given radius of curvature r_0 . The curious parenthetical arrangement of Eq. (19) is intended to aid in the demonstration that, when expressed in term of spin tunes, the “potency’s” of magnetic and electrically induced MDM precessions are in the same ratio as the bending potencies.

Co-magnetometry

This section, which discusses spin dependence, is superfluous, in the sense that there is little or no available data concerning cosmic ray spin dependence. It is included to advertise the importance of spin measurement in laboratory-based storage rings, for the precise determination of resonance energy widths, and to make the point that spin dependence could, in principle, enter into the interpretation of cosmic ray observations.

For particles at rest “co-magnetometry” in low energy “table-top particle traps” has been essential; especially for the *direct* measurement of anomalous magnetic dipole moments (MDMs), storage ring technology with beam pairs that can counter-circulate simultaneously in a storage ring with superimposed electric and magnetic bending is required. In this context the term “mutual co-magnetometry” can be used to apply to “beam type pairings” for which both beams have frozen spins.

In an idealized electromagnetic storage ring, the fields are “cylindrical” electric $\mathbf{E} = -E_0\hat{\mathbf{x}}r_0/r$ and, superimposed, uniform magnetic $\mathbf{B} = B_0\hat{\mathbf{y}}$. The bend radius is $r_0 > 0$. Terminology is useful to specify the relative polarities of electric and magnetic bending: Cases in which both forces cause bending in the same sense will be called “constructive” or “frugal”; Cases in which the electric and magnetic forces subtract will be referred to as “destructive” or “extravagant”.

There is justification for the “frugal/extravagant” terminology. Electric bending is notoriously weak (compared to magnetic bending) and iron-free (required to avoid hysteretic effects) magnetic bending is also notoriously weak. As a result, an otherwise-satisfactory laboratory configuration can be too “extravagant” to be experimentally feasible.

For a particle with spin circulating in a (horizontal) planar magnetic storage ring, its spin axis precesses around a vertical axis at a rate proportional to the particle’s anomalous magnetic dipole moment, $\mathcal{G}$. For an “ideal Dirac particle” (meaning $\mathcal{G} = 0$) in a *purely magnetic field* the spin precesses at the same rate as the momentum—pointing always forward for example. Conventionally the spin vector’s orientation is specified by the in-plane angle α between the spin vector $\mathbf{S}$ and the particle’s momentum vector $\mathbf{p}$ (which is tangential, by definition). For such a “not-anomalous” particle the spin-tune Q_M (defined to be the number of 2π spin revolutions per particle revolution) therefore vanishes, in spite of the fact that, in the laboratory, the spin axis has actually precessed by close to 2π each turn.

In general, particles are not ideal; the directions of their spin vectors deviate at a rate proportional to their anomalous magnetic moments, $\mathcal{G}$, and their spin tunes differ from zero even in a uniform magnetic field. Note also, that a laboratory electric field produces a magnetic field in the particle rest frame, so a particle in an all-electric storage ring also has, in general, a non-vanishing spin tune Q_E . Along with $\mathcal{G}$ and Q , all of these comments apply equally to the polarization vector of an entire bunch of polarized circulating particles.

By convention, in the Bargmann, Michel, and Telegdi, BMT-formalism, the orientation of the spin vector $\mathbf{S}'$ is defined and tracked in the rest frame of the circulating particle, while the

electric and magnetic field vectors are expressed in the lab. The spin equation of motion with angular velocity $\mathbf{\Omega}$ is

$$\frac{d\mathbf{S}'}{dt} = \mathbf{\Omega} \times \mathbf{S}', \quad (22)$$

with orbit in the horizontal (x, z) plane assumed, where

$$\begin{aligned} \mathbf{\Omega} &= -\frac{q}{\gamma mc} \left((\mathcal{G}\gamma) cB_0 + \left(\left(\mathcal{G} - \frac{1}{\gamma^2 - 1} \right) \gamma \beta^2 \right) \frac{E_0}{\beta} \right) \hat{\mathbf{y}} \\ &\equiv -\frac{q}{\gamma mc} \left((Q_M) cB_0 + (Q_E) E_0 / \beta \right) \hat{\mathbf{y}}, \end{aligned} \quad (23)$$

This equation serves to determine the “spin tune”, which is defined to be the variation rate per turn of α , as a fraction of 2π . Spin tunes in purely electric and purely magnetic rings are given by

$$Q_E = \mathcal{G}\gamma - \frac{\mathcal{G} + 1}{\gamma}, \quad Q_M = \mathcal{G}\gamma, \quad (24)$$

where γ is the usual relativistic factor. Note that the sign of Q_M is the same as the sign of $\mathcal{G}$, which is positive for protons—proton spins precess more rapidly than their momenta in magnetic fields. Deuteron spins, with $\mathcal{G}$ negative, lag their momenta in magnetic fields. With $\mathcal{G}$ positive, Q_E increases from -1 at zero velocity, eventually switching sign at the “magic” velocity where the spins in an all-electric ring are “globally frozen” relative to the beam direction. When a particle spin has precessed through 2π in the rest frame it has also completed one full revolution cycle from a laboratory point of view; so the spin-tune is a frame invariant quantity.

In a celestial storage ring with predominantly gravitational bending, to represent a physical limitation on the magnetic bending force at radius r_0 , there can be a go, no-go condition such as

$$|\eta_M / \eta_G| < 1/3.$$

The resulting magnetic force dependence on direction causes an $\eta_M > 0$ (call this “constructive”) or $\eta_M < 0$ (“destructive”) perturbation to shift opposite direction orbit velocities (v) of the same radius, one up in radius and one down, resulting in two stable orbits in each direction. For stored beams, any further $\Delta\eta_M \neq 0$ change causes beam velocities to ramp up in kinetic energy ($KE = \mathcal{E} - mc^2$) in one direction, down in the other.⁴

Depending on the sign of magnetic field B , either the lighter or the heavier particle bunches can be faster, “lapping” the slower bunches periodically, and enabling “rear-end” nuclear collision events. (The only longitudinal complication introduced by dual beam operation in laboratory storage rings is that the “second” beam needs to be injected with accurate velocity, directly into stable RF buckets.)

3 Part 1: “Sun accelerator”, galactic cosmic ray injection

“Start up” and “topping up” solar injection

As an accelerator, the sun needs to have an injector, at least for “start up”. Commonly, the injector is also available for “topping up”. As conceived in this paper, only the first capability is required. The sun itself can replenish intermediate energy particles from its own cosmic ray “atmosphere”.

There seems to be no credible direct way in which particles, say multi-MeV scale protons, can be accelerated from the solar wind up to the GeV energy range. However, once started up, the solar accelerator is self-sufficient, until it is not; for example once every 11 years, while the sign of magnetic dipole component is reversing. Within these periods a self-sufficient cosmic ray atmosphere will have been established, which probably has enough “inertia” to survive the reversal.

⁴In this paper there are several different types of “energy”. The most important of these is the total relativistic particle energy $\mathcal{E}$, with font chosen to be calligraphic, in order to avoid clashing with E , the symbol for electric field.

Arbitrary nuclear particle type A notation

To simplify the formulas, and to emphasize that the dominant centripetal force is gravitational, and proportional to the nuclear mass, we will use A , as in $A = Z + N$, as subscript for (nuclear) particle 2. This means, to adequate precision, that $m_A = Am_p$ or that the rest mass of particle-2, is A , when measured in GeV units. To simplify the treatment of ultra high energy performance we are generalizing the formalism to a single nuclear particle for each value of mass index, namely $A = Z + N$. In other words, there is a unique particle type A for any integer value of A , in spite of the fact that particle names retain their traditional identification with Z ; e.g. for beryllium Be, implies that $A = 9$ and $Z = 4$.

There are grounds, based on electrodynamics and general relativity, for excluding photons from the class of “particles” that can follow orbits under consideration.^{5 6} However, electrons and positrons, are covered, even though their small masses make them more responsive to magnetic fields.

The radius r_A of a unique circular orbit around the center of the sun for a particle A , with relativistic gamma-factor γ_2 , a standard radius value is defined by

$$r_{A-\text{sun}-\text{lim}_1} = \frac{G M_{\text{sun}}}{2c^2} = 738.24 \times \gamma_2 \text{ m.} \quad (25)$$

This value is less than the actual radius of the sun for $\gamma_2 = 1$, since it assumes the entire mass of the sun to be at its center. This result no longer depends on the mass of (nuclear) particle 2. Such an orbit would be internal to the sun, making it unphysical. It can be compared numerically with the radius of the sun, which is $R_{\text{sun}} = 0.6957 \times 10^9 \text{ m}$. To find the smallest particle energy consistent with circulation outside the sun, based entirely on the sun’s gravitational attraction, the ratio of these two results yields

$$\gamma_{2-\text{sun}-\text{lim}} = \frac{R_{\text{sun}}}{r_{A-\text{sun}-\text{lim}_1}} = \frac{0.6957 \times 10^9}{738.24} = 0.943 \times 10^6. \quad (26)$$

By chance, since the rest mass of a proton is 0.938 GeV, this means that only protons with energy greater than 10^6 GeV can circulate stably in a circular gravitational orbit around the sun, without the possible aid of some (presumably magnetic) centrifugal, repulsive radial force pointing away from the sun.

Who ordered cosmic ray positrons and anti-protons?

According to the data in Figure 5, there are (negative) anti-proton cosmic rays present in the solar system. They are not abundant, but even one confirmed anti-proton would need to be explained. Certainly, if protons can be accelerated clock-wise (CW) then anti-protons can be accelerated CCW. The same goes for electrons and positrons. In short, the sun would then be a colliding beam storage ring. This would be doubly welcome, as it would help to answer the question “Who ordered $\bar{p}$ and e^+ cosmic rays?”.

The relative relative abundances of p and $\bar{p}$ ’s, on the one hand. and e^- and e^+ ’s on the other, have now been measured. Along with careful calculation of the most likely QED production channels, checking these abundance ratios would provide luminosity-independent consistency checks of the proposed solar acceleration mechanism.

By referring to Figure 11, one sees that the highest energy cosmic ray protons just barely fail to meet $\gamma_{2-\text{sun}-\text{lim}}$ condition evaluated in Eq. (26). The data shows that all other cosmic ray nuclear isotopes fail, but also just barely, to meet the same condition. This statement depends on the abscissa label of the figure, which is “Kinetic Energy Per Nucleus [GeV]”.

⁵This comment is based on conversation with Saul Teukolsky. With the photon mass being exactly zero, we are saved from the embarrassment of establishing the value of the $\gamma_2/m_{\text{photon}}$ ratio in the vanishing mass limit.

⁶With $G_n = 6.67258 \text{ m}^3 \text{ s}^{-2} \text{ kg}^{-1}$, $M_{\text{sun}} = 1.989 \times 10^{30} \text{ kg}$, and conversion factor from Joule to GeV = 6.242×10^9 , the limiting radius is $r_{2-\text{sun}-\text{lim}} = 1.0631 \times 10^{10} \text{ m}$. For the six proton energies plotted in Figure 16, the limiting radius values, in units of 10^{10} meters, are 8.085, 1.973, 1.518, 1.366, 1.269, and 1.177.

But it is more important that there must be a mechanism in nature capable of accelerating nuclei to, say, 10^5 GeV/nucleus, in the context of the present paper. To meet this “start-up” condition one can, or must, assume that some particles with energies in excess of 10^5 GeV/nucleus are entering the solar system from the solar galaxy, depending upon the magnetic dipole field, shown, for example, in Figure2.

There is some leeway here, in that the start-up gravitational centripetal force has been neglected, based on a radius ten times the solar radius, and the centripetal gravitational force varies with inverse square law dependence on radial distance.

The Alfven-surface

There is another issue influencing the sun as accelerator. It concerns the Alfven sphere, with radius

$$R_{\text{Alfven}} \approx 10 \times R_{\text{sun}}, \quad (27)$$

or 4% of the distance to the earth; a characteristic radius of the magnetohydrodynamic field surrounding the sun. See Figure (4), copied from S.R. Cranmer, et al.[4], shows features at elevations from the sun’s surface out to 100 times the sun’s surface, which is closely related to the range of nuclear and electron orbit radii being discussed in the present paper.

Copying from that paper,

“It is known that the strongest forces acting on the plasma undergo a transition from being mostly magnetic, near the Sun, to mostly hydrodynamic – i.e. depending on gas-pressure gradients and nonlinear inertial gas flow terms – far from the Sun. A common way to quantify this transition is to locate the Alfven surface, the place where the radially increasing solar-wind speed $[u]$ exceeds the radially decreasing Alfven speed $[V_A]$. “. One sees from the figure that the Alfven radius (though significantly variable as a function of time) is approximately 10 times the radius of the sun.

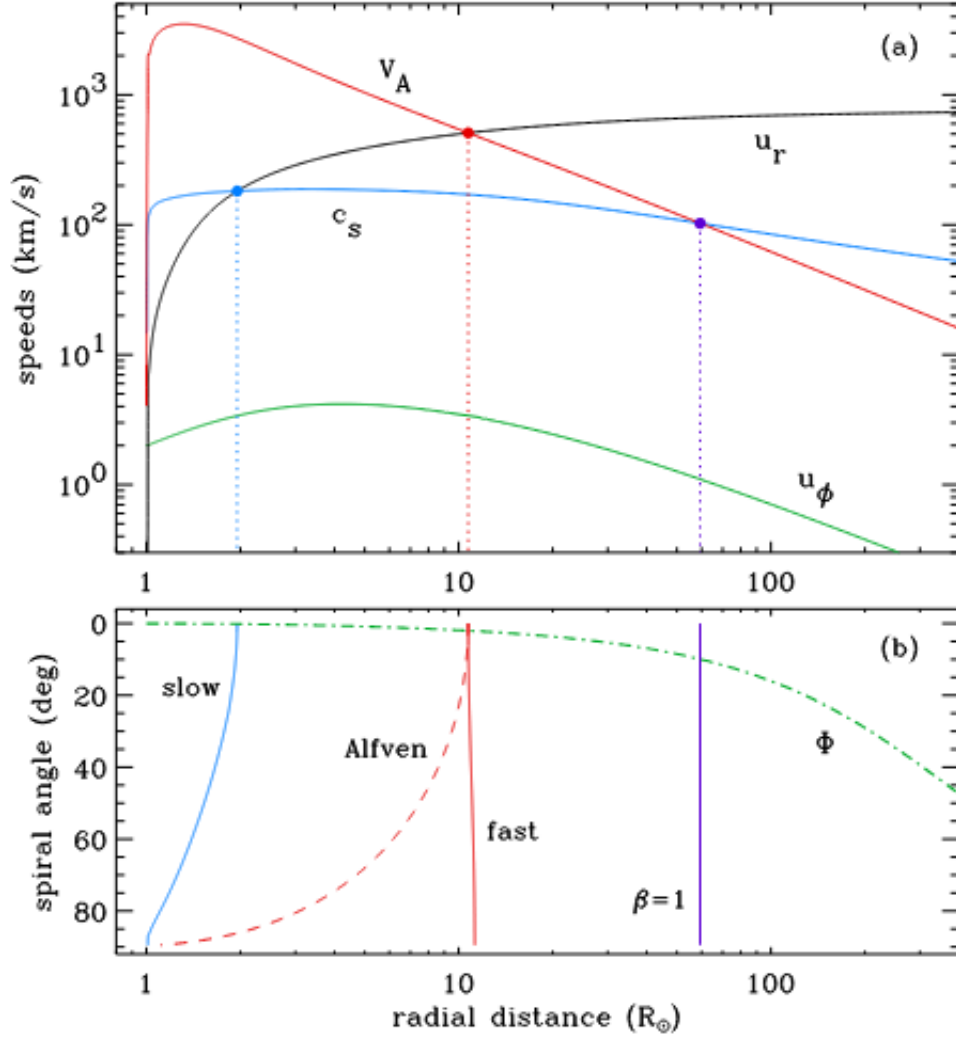
NASA’s Parker Solar Probe (PSP) encountered the specific magnetic and particle conditions at 18.8 solar radii that indicated that it had penetrated the Alfven surface; Averaging these figures, one expects a value r_{Alfven} in the range of 10 to 20 times R_{sun} for the Alfven radius. This will become significant in the context of discussing particle injection onto the sun as accelerator, for which the Alfven physics will be important. But the existence of stable circular orbits around the sun is only significant if there is an efficient injection mechanism, and the Alfven physics will be germane to this paper.

Understanding of the solar wind has been expanding continuously ever since its most clear elucidation by Parker 70 years ago. The abstract to a 2023 paper by Crammer et al.[5] reads as follows

“ The solar wind is the extension of the Sun’s hot and ionized corona, and it exists in a state of continuous expansion into interplanetary space. The radial distance at which the wind’s outflow speed exceeds the phase speed of Alfvénic and fast-mode magneto-hydrodynamic (MHD) waves is called the Alfven radius. In one-dimensional models, this is a singular point beyond which most fluctuations in the plasma and magnetic field cannot propagate back down to the Sun. In the multi-dimensional solar wind, this point can occur at different distances along an irregularly shaped “Alfven surface....Combined with recent perihelia of Parker Solar Probe, these studies seem to indicate that the Alfven surface spends most of its time at heliocentric distances between about 10 and 20 solar radii. It is becoming apparent that this region of the heliosphere is sufficiently turbulent that there often exist multiple (stochastic and time-dependent) crossings of the Alfven surface along any radial ray. Thus, in many contexts, it is more useful to make use of ... a complex “Alfven zone” rather than one closed surface.

“

Figure 4 displays a combination of theoretical behaviors in the “Alfven zone”. As well as in the original caption, this figure is explained in the new caption. There is hardly anything in the two captions to Figure 4 that is not relevant for the present paper. Here we emphasize the “(stochastic and time-dependent)” characterization of the region which will be referred to as the “Alfven zone” in the present paper.



Example one-dimensional solar-wind model, with quantities plotted vs. heliocentric radial distance r , in units of solar radii $R_{\odot}$: (a) radial outflow speed (*black*), sound speed (*blue*), Alfvén speed $V_{A,r} = B_r/(4\pi\rho)^{1/2}$ (*red*), and azimuthal flow speed (*green*); (b) locations of MHD critical points shown with respect to both r and the Parker spiral angle Φ (*red and blue curves*), plotted alongside the self-consistent $\Phi(r)$ for the Weber and Davis (1967) model (*green dot-dashed curve*). In both panels, *purple* denotes the $\beta = 1$ surface (see text).

Figure 4: Figure copied with original caption from reference[5], showing, in particular, the radius of the Alfvén surface, located at approximately 10 times the solar radius. Note: η in this figure is not velocity as fraction of the speed of light.

Every solar cosmic ray has to make its way through the Alfvén zone frequently and periodically from its birth to its death. It is only these passages which can contribute stochastic noise to the solar cosmic ray equilibrium. Its evolution outside this region, to be described below, is governed by non-stochastic Hamilton-Jacobi evolution. It remains hard to understand how all particle types, hadrons, leptons, and their anti-particles, can evolve identically in the fully-relativistic limit. But, at least, it is clear that this behavior is governed by passages through the Alfvén region of the sun.

Magnetic bending at star and planet equators

The magnetic dipole moments common to most stars and planets provide a magnetic field with magnetic dipole pattern. The magnetic field to be emphasized in this section is aligned with the rotational axis and uniform close to the equatorial plane. By symmetry, external to the sun, this field parallel to the axis of rotation of the sun is sufficiently uniform for large aperture

treatment. The “natural” cyclotron orbits are circles of radius greater than the solar radius, and centered on the axis in the equatorial plane. Any cosmic ray orbit discussed in this paper follows a helix of increasing or decreasing radius, slowly drifting “out of plane”, or “vertically”, in accelerator terminology, away from, or toward the equatorial plane.

The Parker solar wind provides another essential field component. It is the latitudinal (or “tangential” in Frenet-Serret accelerator terminology) component of electric field, shown in the lower left corner of Figure 3. Though small at the surface of the sun, this electric field, which is caused by rotation of the sun’s magnetic moment, becomes substantial at the surface labeled “effective radial magnetic field source (ERMFS)” in the figure. This tangential electric field can serve the same role as the RF cavities in terrestrial circular accelerators or storage rings.

When referring to a star or planet as an accelerator, these circles are slowly expanding or contracting, nearly-closed orbits. Since free space is a fairly good vacuum, and only a quite small number of turns expected, no vacuum chamber is required. In a linearized sense then, the ring aperture is the infinite half-space external to this sphere; or, rather, to the nonlinear dynamically limited portion of this half-space. Unlike any functioning terrestrial accelerator, there is no significant focusing, other than the weak geometric focusing of closed (except for precession) elliptical orbits. As a consequence nonlinear bending will surely be unimportant.

To make this kind of celestial accelerator better yet one could wish for focusing to better preserve beam emittances; otherwise known as “Courant invariants”. These were named after “Ernest”, who was the son of “Richard”, co-author of “Courant and Hilbert”, with the “Hilbert” of Hilbert space fame. Their colleagues in Gottingen, Germany, include far too many to list famous physicists and mathematicians, including Riemann in 1859.

Helical Alfvén guiding centers could supply such focusing if they lay in the equatorial plane. But the only important Alfvén guiding centers are aligned with magnetic fields perpendicular to the equatorial plane. Any useful net beam acceleration must, therefore, be accomplished in a fairly small number of turns, or possibly just a fraction of one turn.

Though some protons are constantly being lost, they remain predominant, since they are constantly being regenerated by nuclear collisions. Note, though, that it is only low energy protons, not high energy cosmic rays that are being regenerated.

Definitions of “magnetic rigidity” and “fine tuning” complications

For interpreting graphs published by International Space Station (ISS) authors it is important to understand “magnetic rigidity”.

A universal nuclear mass m_U can be defined as the mass of a carbon-12 nucleus, divided by 12. To express the mass of a nuclear isotope in the form Am_U the mass of Z electrons must have been subtracted from the atomic mass. In accelerator physics jargon, the definition of “magnetic rigidity” of a nuclear isotope (Z,A) species, on a circular arc of radius ρ in magnetic field B is referred to as “B-rho” where, in MKS units,

$$B\rho = \text{magnetic rigidity} = (A/Z)m_U\gamma\beta c/e. \quad (28)$$

We are mainly interested in order of magnitude precision, limiting discussion to protons, for which $(Z, N, A)=(1,0,1)$, and “ α -material”, for which $(Z, N, A)=(Z, N=Z, A=2Z)$.⁷ This terminology may seem to imply that any α -material nucleus can, for some purposes, be treated the same as any other, irrespective of their particular Z -values. This is intentionally and precisely what is being implied.

Specific to magnetic bending; all such isotopes have the same curvature while in the same magnetic field. Within the Parker solar wind model, all such particle orbits have the same curvature while in the same magnetic field.

Also, a standard mass m_U is defined for which, quoted as an energy, $m_Uc^2 = 1 \text{ GeV}$. Also, the proton rest energy is 0.938 GeV , not very different from m_U .

⁷The “ α -material” approximation is accurate for a few low Z cases, but becomes increasingly less valid as N exceeds Z above helium, carbon and oxygen and beyond.

In many papers describing results from the ISS and elsewhere, a parameter also referred to as “rigidity” is defined as

$$R = (pc)/(Ze) = B r_L, \quad (29)$$

which differs from what accelerators refer to as “magnetic rigidity”. Evidently, ρ and r_L are two names for the same thing. Also, if, one refers to the ISS quantity as “rigidity per charge” in units of eV, as a momentum in units of GV, one obtains the same result as an energy in units of GeV in accelerator physics terminology.

The radius of the sun is approximately 696,340 kilometers or 0.6963×10^9 m. For an example needed later in the present paper, one can determine the energy in GeV for a proton traveling along a circular orbit at a (minimal) value of the Alfven radius distance from the center of the sun given by $\rho = 0.7 \times 10^{10}$ m, or approximately 10 solar radii, for a (maximal value) magnetic field of 10 gauss, i.e. 0.001 T.

Eq. (28) and Eq. (29, after appropriate substitution of symbols, become

$$\rho = \frac{(m_U c \gamma_p \beta_p)}{B e} \quad \text{or, for proton momentum, } p_p c/e, \quad \frac{p_p c}{e} = B \rho. \quad (30)$$

One sees, from the units, that what is actually being determined by “B-rho” is the particle momentum, in this case p_p , for the proton, expressed in convenient units, in a formula that is valid both relativistically and classically. This is the reason why the abscissa horizontal label of “energy dependence” in ISS is correctly expressed in GV units, over the full range of energies. In the fully relativistic regime GV and GeV can be used interchangeably, for a particle having charge equal in magnitude to the ccharge of an electron.

Evaluating the final equation in Eq. (30), the value of the B-rho product is $0.001 \times 0.7 \times 10^{11} = 0.7 \times 10^8 \text{ eV}$, or 0.07 GeV; this is the energy of a proton, steered by magnetic force only, in circular motion along the Alfven radius of the sun, which is approximately 0.1 AU.. Since the rest energy of the proton is approximately 1 GeV, *this proton would be non-relativistic*, with $\gamma_2 = \gamma_p \approx 1$.

This result can be compared with Eq. (9)) which gave 10 keV as the radius of a gravitationally-imposed, circular proton orbit along the sun’s radius. Comparing these results, one sees that the magnetic force on the 0.07 GeV proton, even though non-reletavistic itself, would dominate over its gravitational attraction toward the sun.

On the other hand, comparing this result with, say, the experimental data shown below in Figure (16), the “pseudo-orbits” shown in that figure, having enegies in the 1 to 10 GeV range, are fully relativistic, and the higher energy orbits must, therefore, be predominantly gravitationally imposed.

However, the magnetic field at the Alfven radius varies significantly, ranging from 1 gauss to 10 gauss, to some extent correlated with the 22 year periodicity of the sinusoidal solar magnetic field. Neglecting the gravitational force of the sun on the proton, (which is substantially smaller) we have calculated the proton energy for a proton circulating around the sun, at a radius ten or twenty times the radius of the sun to be in a range from 1 to 10 GeV.

It is only because the start-up bending is magnetic, that positive and negative particles can counter-circulate in the same ring. Later, both beams have been sufficiently accelerated for the magnetic bending to “saturate”, which enables the bending to be gravity-dominated, without losing either of the counter-circulating beams. If true, this is a remarkable instance of “fine-tuning” in nature.

The data in Figure 11 shows ISS-measured cosmic rays for energies per nucleus from 10^1 to more than 10^5 GeV/nucleus. The gravitational force of the sun is proportional to $1/rho^2$ and the centripetal force needed to hold high momentum particles in circular orbit is proportional to $1/rho$. The radius of the solar system is not less than 10^5 AU, which is six orders of magnitude greater than the Alfven radius of the sun. It seems, therefore, that cosmic rays of energy as high as 10^{11} GeV can be held captive in the solar system.

G&M solar bending and electric acceleration

The sun's magnetic dipole moment field magnetic field, alone, given by Eq.(1), and normal to the sun's equatorial plane, is capable of bending low energy nuclear isotopes and electron's into counter-circulating orbits above the equator of the sun. During an "injection phase" the kinetic energies of all of these particles are being accelerated by the "longitudinal" electric field component shown, for example in Figure 3.

The actual situation is further complicated by magnetic forces, which "saturate" as the speed approaches the speed of light. While still non-relativistic, the magnetic deflection strengths track proportionally with increasing momentum. Once a particle's speed has become essentially equal to the speed of light, it is only the gravitational force that can provide the bending force required for further increase in momentum. One has to anticipate an intermediate situation during which the magnetic bending has become negligible. This might be referred to as the end of the injection phase.

Later, for cosmic ray particles remote from the sun, the magnetic bending may eventually becomes significant again. Though weak, this magnetic bending will "corrupt" the otherwise predictable relativistic Keplerian cosmic ray particle trajectories. To the extent these weak magnetic fields are known, they can be treated as perturbations of ideal Hamilton-Jacobi-Kepler orbits.

Fortunately, there is splendid experimental ISS-AMS data bearing on the relative importance of magnetic and gravitational bending of cosmic rays. Shown in Figure 5, are energy spectra, expressed as rigidities in GV units, for protons, electrons, positrons and anti-protons.⁸ For the superposition of magnetic and gravitational bending we need to derive the analogs of Eqs. (20) which apply to the superposition of magnetic and electric bending. A gravitational/magnetic Lorentz force law analog is,

$$\mathbf{F}_{GM}(r) = -(Gm_1)\frac{\gamma_2 m_2}{r^2}\hat{\mathbf{r}} + qc\hat{\boldsymbol{\beta}}\hat{\mathbf{z}} \times B(r)\hat{\mathbf{y}} \quad (31)$$

$$= \left((Gm_1)\frac{\gamma_2 m_2}{r^2} + q\beta cB(r) \right) (-\hat{r}), \quad (32)$$

where $\hat{\mathbf{x}} = \hat{\mathbf{r}}$, $\hat{\mathbf{y}} = \hat{\mathbf{B}}$ and $\hat{\mathbf{z}} = \hat{\mathbf{v}}$, form a right-handed triplet of mutually perpendicular unit vectors.⁹ In the gravitational term the product (Gm_1) has been combined; the reason for this is that, while G itself, is poorly known, the product of the Sun's mass M_{sun} and the gravitational constant G is known with value known to 10 decimal places, which, for our purposes can be expressed adequately as

$$(GM_{sun}) \approx 1.3271244 \times 10^{+20} \text{ m}^3 \text{ s}^{-2}.$$

This product is known as the "Sun's standard gravitational parameter"; it has a universal symbol which will not be needed, and is therefore not shown. Note that, with these physical dimensions, division by a velocity squared, such as c^2 , produces a length in meters.

As already stated more than once, particle 1 is heavy, particle 2 light. Note also that, in our units, $m_2 c^2$ is the rest energy in GeV.

We wish to mimic the η_E and η_M partitioning defined in Eq. (20). Unfortunately, in the G&M case we are not at liberty to choose the magnitudes of either the gravitational or magnetic force, nor the sign of the magnetic field; they are provided by nature.

We therefore define η_G^{**} and η_B^{**} , where overhead ** or * symbols provide warning that the fractions do not sum to 1. Bending fractions are then defined by.

$$\eta_G^{**} = (Gm_1)\frac{\gamma_2 m_2}{r^2}, \quad \eta_M^{**} = q\beta cB(r). \quad (33)$$

Unlike a laboratory storage ring, which has a fixed bending radius, an astronomical accelerator has no such constrained radius. However, like a laboratory-based E&M ring, beams of a given

⁸GV units are appropriate for momentum pc , or rather for "momentum by charge", where it is convenient to describe the charge as Z, in units of the proton charge e, with the numerical value of e being 1.

⁹As elsewhere in this paper, the factor c in Eq. (32) is superfluous, since its value is 1; its presence serves as reminder concerning the velocity dependence of the Lorentz force law.

type, say protons, can rotate either CW or CCW. If there are also anti-protons, i.e. $\bar{p}$ -type, the gravitational force will be the same for both beam types, but the magnetic forces will be opposite, constructive for one type, destructive for the other.

A priori, we had no reason to expect to see $\bar{p}$'s at all. Yet the AMS detector results in Figure 5 show, roughly, one $\bar{p}$ for every 10^5 protons. Clearly the p 's and $\bar{p}$'s are traveling stably in opposite directions. This means that, for nuclear particles, the gravitational bending is predominant in both cases.

We also had no *A priori* reason to expect to see electrons, let alone positrons. The AMS detector shows, roughly, one positron for every 50 electrons, and one anti-proton for every ten thousand protons, though these ratios only stabilize in the fully relativistic regime.

Clearly the electron's and protons need to be streaming initially in opposite directions. This means that, even for light (i.e. lepton) charged particles, the magnetic bending is more important for electrons than for protons.

In passing one notes that electrons and protons traveling in opposite directions around the sun turn the sun into a “colliding beam storage ring”. This, no doubt, is how positron's and anti-protons enter the picture. Since this physics will be trivial once the cosmic ray generation is understood semi-quantitatively, this mechanism is not discussed further in the present paper.

In short, Figure 5, suggests that all charged particles are being accelerated, presumably while temporarily “captured” in a gravitational storage ring. Furthermore, there is ample data provided by the same figure to establish the relative strength of the magnetic and gravitational bending forces more quantitatively. I, personally, can see no reason to doubt that the acceleration in question is provided by the external fields, gravitational and magnetic, of the sun. For nomenclature one can express the sun as a G&m ring, meaning that the bending is predominantly gravitational.

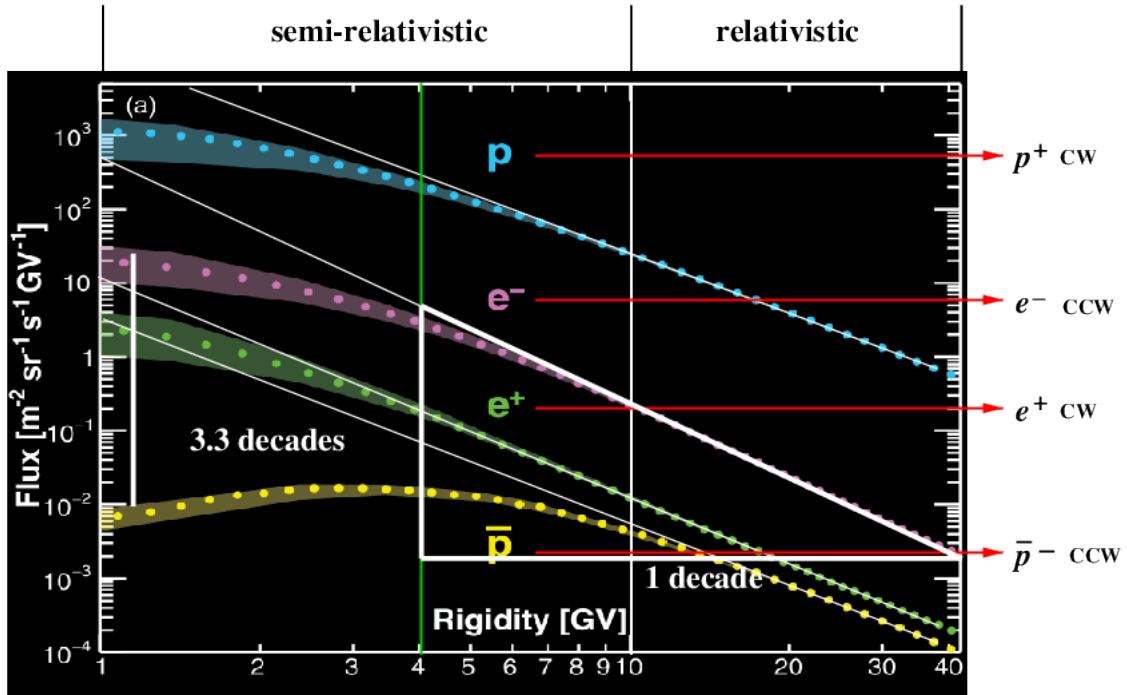


Figure 5: Figures copied from reference[6], showing rigidity dependence (also known as momentum over charge dependence) and relative abundances of protons, electrons, positrons, and anti-protons measured by the AMS detector in the ISS, interpreted as four separated beams of protons, electrons and their anti-particle being accelerated by the sun, with superimposed gravitational and magnetic bending, and Parker electric field acceleration. For the asymptotic extrapolation, $\text{Flux}[\text{db}] = -10 \log_{10} 10^{3.3} = -33 \text{ db/dec}$. The protons have become fully relativistic only above 10 GeV, while the electrons have become fully relativistic midway through the region labeled “semi-relativistic”.

4 Correlation: cosmic ray intensity with solar oscillation phase

Magnetic field imposed complications

The primary importance of magnetic fields in the solar system is not inside the sun or planets; it is in the free space external to these bodies. There is a strong tendency for massive atoms to fall back into the sun or one of the planets. In equilibrium, it is predominantly protons, electrons, and alpha material (predominantly α -particles and deuterons) that continue to circulate freely and longer, in more or less isotropic directions. Meanwhile electrons and nuclear particles will have merged into atoms that are captured in the atmospheres of planets.

It is predominantly higher momentum protons, deuterons and α -particles that continue to circulate freely. In equilibrium these particles will have gravitated toward the most massive objects in the solar system, namely either the sun or Jupiter.

Quoting from reference [7], “NASA’s Juno mission to Jupiter made the first definitive detection beyond our world of an internal magnetic field that changes over time, a phenomenon called secular variation. Juno determined the gas giant’s secular variation is most likely driven by the planet’s deep atmospheric winds.”[7] Also “The field rotates with the approximately 9 hour rotational period of the planet.”.

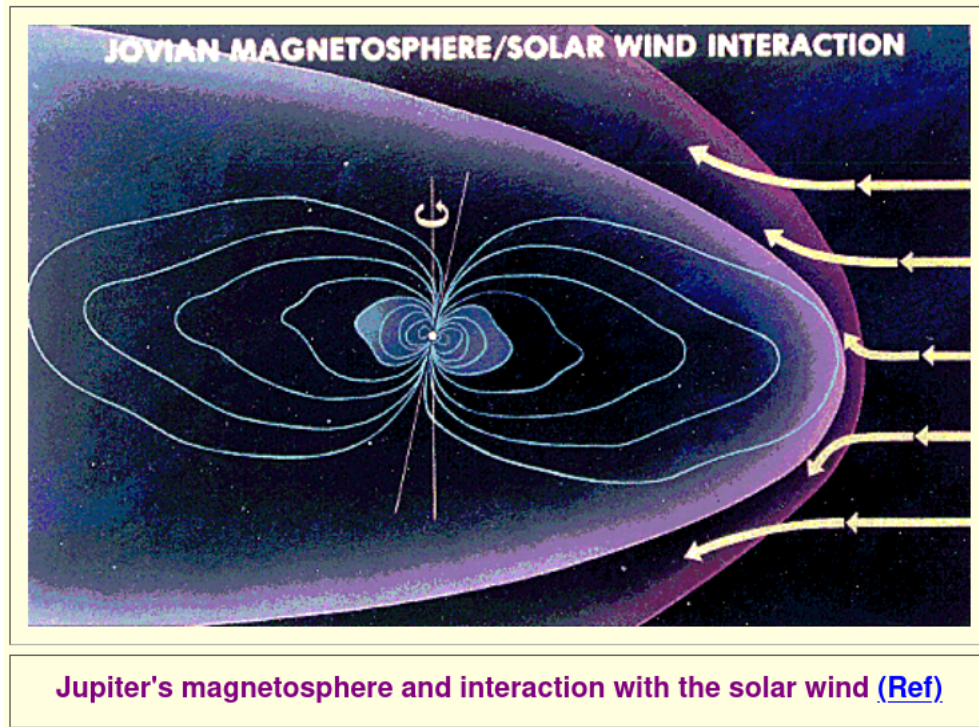


Figure 6: Jupiter magnetic field contours, copied from reference [7].

When produced in standard terrestrial alpha radioactive decay, alpha particles generally have a kinetic energy of about 5 MeV and a velocity in the vicinity of $0.04c$. At this speed the time of flight of an alpha particle from Jupiter to the sun is approximately $0.0007/0.04 = 0.02$ years, or about 7 days. At higher energy or lower mass this time would, of course, be less. the presence of magnetic fields in

By way of contrast, with the diameter of the solar system being about one light year, a fully-relativistic particle traveling to the sun and back (presumably following a highly eccentric elliptical orbit) would have taken a year or so to get to the sun and back. This means, for example, that the sun’s magnetic fields may have reversed, with ten percent probability, during the longest possible duration of a solar-based cosmic ray particle. For purposes of this paper, this is taken to mean, even though the cosmic ray intensities are correlated with the sun’s orientation reversal, that the cosmic ray acceleration process does not need to be restarted every 11 years. Though important cosmic ray features may vary in phase with the sun’s magnetic polarity, the

cosmic ray “atmosphere” and the solar wind are less in lock step with each other than is true for a laboratory accelerator being refilled from its injector.

Nevertheless, there remains a strong correlation between cosmic ray intensity with the phase of the sun’s 22 year periodic variation. *This correlation is plausibly explained in the “standard model” of extra-galactic cosmic ray production, by the deflection of remotely generated cosmic rays being caused by solar magnetic fields they encounter, after they have entered the solar system.*

Understanding this correlation is also challenging for a theory assuming that cosmic rays are purely of solar origin. It is hard to understand the mechanism which keeps the cosmic ray “atmosphere” synchronized with the solar wind. Without such synchronism it is hard to understand how cosmic ray flux intensities can be steadily replenished.

One is tempted to assume that no such synchronism exists. This would mean, however that partially accelerated cosmic rays would be as likely to return to the sun “out of phase”, meaning they would be decelerated.

The problems are of very different character, however. Nevertheless. the presence of substantial magnetic fields in the free space within the solar system seriously disrupts any simple, periodic, closed form theory of solar cosmic ray production.

All this makes it important to learn what is known about the correlation between cosmic ray intensity correlation with phase of the sun’s 22 year oscillation period. The “open solar flux” (OSF) represents the variable heliospheric magnetic flux. The “galactic cosmic ray” (GCR) measure represents cosmic ray intensity near Earth. Quoting Koldobsky[8], “The flux of *galactic* cosmic rays (GCRs) outside the heliosphere is generally assumed to be constant at time scales shorter than a hundred thousand years.” as well as “The GCR flux variability is known to be delayed with re-respect to solar activity, leading to the so-called “hysteresis” effect of the phase shifts in the development of the 11-year solar cycle in both indices. Here the *galactic* qualifier has been emphasized to acknowledge that correlation between cosmic ray intensity variation presents serious problems for both solar-based and galactic-based cosmic ray generation.

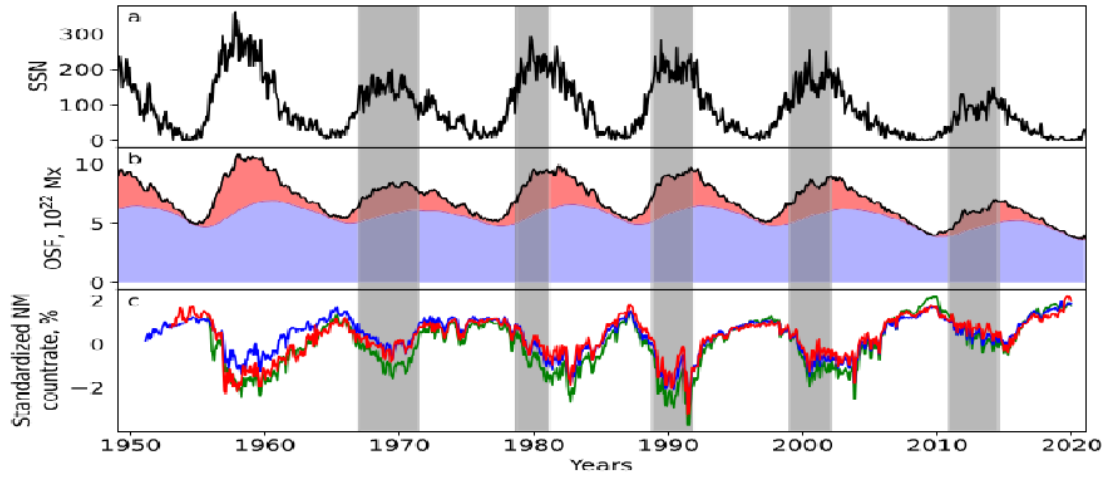
The long-term variability of GCRs is routinely monitored by a global network of ground-based neutron monitors (NMs) that are sensitive to the nucleic component of the cosmic-ray-induced atmospheric cascades and located, as a worldwide network, around the globe since the 1950s.

Again quoting Koldobsky[8], “All of these three indices appear highly coherent at a timescale longer than a few years with persistent high coherence at the timescale of the 11-year solar cycle. The GCR variability is delayed with respect to the inverted SSN (solar sunspot number) by about eight 27-day Bartels [a measure of solar activity rotation cycles] is 1/2 of a year on average, but the delay varies greatly with the 22-year cycle, being shorter or longer around positive A+ or negative A-.”

Time Lag measures of SSN, OSF, and NM from 1950 to 2020 are platted in Figure 7, along with the original caption. Frequency compositions are commented upon in the new caption. When produced in standard terrestrial alpha radioactive decay, alpha particles generally have a kinetic energy of about 5 MeV and a velocity in the vicinity of $0.04c$. At this speed the time of flight of an alpha particle from Jupiter to the sun is approximately $0.0007/0.04 = 0.02$ years, or about 7 days. At higher energy or lower mass this time would, of course, be less.

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Datasets used here: **(a)** International sunspot number, v.2.0 (SSN). **(b)** Open solar flux (OSF), including its fast (red shading) and slow (blue) components. **(c)** Standardized NM records for the three cutoff rigidity P_c groups: low (green), moderate (blue) and high (red) (see details in Figure 2). All datasets are shown with the Bartels rotation (27 days) averaging. Periods of the reversals of the heliospheric magnetic field polarity are shaded with gray (Thomas, Owens, and Lockwood, 2014; Krainev et al., 2016).

Figure 7: Time Lag comparisons, 1950-2020, of sunspot number (SSN), open solar flux (OSF) and cosmic ray measure (NM). **(a)** Notice that, unlike the upper two solar plots, the cosmic ray measure remains within a ± 2 -percent range. As regards signs, the upper two solar plots remain faithfully in phase. So also do the cosmic ray deviations. But maximum solar flux coincides with reduced cosmic ray rates and minimum solar flux coincides with increased cosmic ray rates. **(b)** But notice also from the top two figures, that the OSF is a superposition of an AC signal of significantly variable amplitude, or different frequency component, and a DC signal. **(c)** In a DC sense, it might be said that, on the average, the OSF and the cosmic ray flux are proportional. **(d)** Though opposite in phase, so also are the SSN's and the cosmic ray deviations.

1950s.

Systematic particle acceleration

Figure 2, shows a toroidal region surrounding the with transverse area half as big as the sun itself, shaped much like the magnetic field of a ground based particle accelerator, such as the LHC.

Even though the main deflecting force is gravitational, provided by the mass of the sun, there is a more or less constant, vertical magnetic field not unlike the field of a weak focusing storage ring. As shown in Figure 3 there is also a “constant” (modulo fluctuations and 22 year period sinusoidal variation) “longitudinal” electric field. Charged particles with velocities directed more or less parallel to the electric field, will be accelerated.

Though much like ground-based storage rings, there are, of course, many features distinguishing the Sun-SR from ordinary storage rings. *The most important of these differences is that, unlike a ground-based accelerator, the bending field does not ramp up to match the increasing momenta of the candidate cosmic ray charged particles.* Less important, the acceleration is being treated as “DC” rather than “AC-RF”. In fact, discussed in Appendix C is a nonlinear plasma process capable of bunching the cosmic ray particles, much as in a laboratory accelerator, while their energies are low. To begin with we need only discuss the acceleration to high energy of single particles being accelerated without bunching.

Our model, therefore, initially treats each isotope particle individually and incoherently. Once the acceleration of a single particle has been understood, it is not as difficult to understand how beam bunching acquired at low energy can be retained all the way up to high energies. The same comment applies to all charged particle types, including leptons and anti-particles. It is

not immediately obvious how anti-particles come into the picture. Certainly there are neither positrons nor anti-protons present initially in the solar wind. Their source should not be hard to understand, once counter-streaming high energy electrons and proton bunched beams have been understood, but this discussion can be deferred until later.)

Concentrating initially on a single particle, say a 100 MeV proton, and plausibly describing its acceleration to, say 10^{10} GeV, it is easy to envisage a large number of such protons being bunched into an arbitrarily long continuous trains of charged particle bunches. It is also easy to visualize such trains, like Amtrak trains, behaving sporadically. This requires the start-up procedure to be repeated frequently and more or less reliably. The duration of even the most successful train could not be longer than 11 years, which is the longest time the sign of the sun's fields remain the same. But this would be a ludicrously too long over-estimate, based on sporadic solar flares, electric storms, and all.

Each individual, initially 100 MeV, proton once accelerated enough to first escape the sun, follows a relativistically valid, highly elongated precessing Kepler orbit which, eventually, but inexorably, brings it back periodically through the sun's atmosphere, presumably in a zone for which R_{Alfven} is the typical radius. Here it interacts strongly with the sun's solar wind, possibly to an extent that the proton is continues onto an orbit much the same as before, except for having increased energy. See, for example, Figure 8.

While completing spiral turns around the sun, at radius approximately equal to the Alfven radius. These semi-circular match highly-elongated, highly-eccentric, precessing, ellipsoidal relativistic orbits having one focus at the center of the sun. Each such orbit includes more-or-less semi-circular orbit injection and extraction segments. Every other incipient cosmic ray orbit is similar. And the radii of the circular segments are close to the so-called "Alfven radius" R_{Alfven} . All this is illustrated in figures 8 and 15. These assumption reduce the problem to mechanics that can be treated by the Hamilton-Jacobi model, as described in Appendix B. During their lengthy periods remote from the sun, orbit wandering due to unknown magnetic fields needs to be treated as stochastic perturbations; for example as described in reference [9].

Our Hamilton-Jacobi model of planetary orbits resembles quite closely the Sommerfeld relativistic refinement of the 1920, pre-quantum mechanics, Bohr theory of atomic structure. It was Jacobi who developed the method of separation of variables in Hamilton's partial differential equation (PDE) to perform a "canonical transformation" to produce momentum variables that are, themselves, constants of the motion. As well as satisfying special relativity, this is the same mathematics as has been used ever since 1925, to solve the Schrödinger equation in quantum mechanics (QM). All that was missing from modern QM at Bohr-Sommerfeld time was the introduction of non-commuting variables and Heisenberg uncertainty.

Anyway, my proposed model of cosmic ray orbits of elementary particles around the sun is much the same as Sommerfeld's model of electrons orbiting a nuclear atom. The first "constant momentum component" in the Jacobi treatment of orbits is the particle energy $\mathcal{E}$. Bohr's quantized atomic energy level model began with his requirement for energy levels to correspond to classical adiabatic invariants. Our application is much the same, even though the physical application is very different.¹⁰

For our purposes the adiabatic invariance of the orbital energy of a single isotope, such as a proton, is an "orbit element" in Lagrangian terminology. This means that its value is decoupled from the three other canonical momenta, which therefore remain constant while the particle energy increases, during its periodic transit through the Alfven zone of the sun. Averaged over days or months an equilibrium solar atmosphere of cosmic rays develops.

The duration of a "long accelerator run" in the Sun-SR cannot exceed 11 years; any established "beam" will not be able to survive this regular reversal of the sun's magnetic field (and with it the reversal of acceleration directions). However the decay lifetime of the solar atmosphere is presumed to be long compared to 11 years. This is consistent with the time spent during acceleration, being long compared to time spent in a more-or-less stable atmosphere of

¹⁰Bohr's application was consistent with his requirement for the frequencies of radiated photons to correspond to the differences of energies between two energy levels, rather than being harmonics of a single energy level.

cosmic rays, many of which had previously been accelerated to quite high energy, though never high enough to escape the solar-system.

As with land-based storage rings, the details of solar storage ring start-up are bound to be quite messy. For example, start-up may be synchronized with unpredictable sun-spot activity. Let us, therefore, skip to a conjectural low energy start-up condition, with specified proton beam of definite energy and current, circulating stably while passing periodically through the Alfvén zone having radius about ten times the radius of the sun.

It is only the “longitudinal” velocity component of circulating cosmic ray particles that is accelerated. Large transverse velocity components will typically cause the fractional acceleration of such particles to be quite limited, especially for large vertical “out-of-plane” velocity component particles. Furthermore, the acceleration will not necessarily always be positive. But this depends on the beam bunching which will favor positive acceleration but, for now, is being neglected.

There will, however, be a substantial difference of injection efficiencies. Particles approach the sun with equal probability of missing to the left or to the right. But the accelerating efficiency will be half and half, depending on their sign. The centrifugal bending to establish this condition is produced by the “destructive” superposition of gravitational and magnetic bending. Being magnetic, this will be effective one way for positive charges, the other for the other sign.

While the velocities remain non-relativistic, the radial magnetic force will scale up proportionally with the particle momentum, supporting the orbit radius to be greater than the radius of the sun.

5 Part 2: The sun as cosmic ray booster accelerator

Balancing magnetic and gravitational force expressions

In SI units, with $B_{\text{sun}}(r)$ being the longitudinal magnetic field component at radius r above the sun’s equator,

$$F_E = qeE_0, F_M = qe\beta c B_{\text{sun}}(r) \text{ and } \tilde{F}_{G,p} = (Gm_1) \frac{\gamma_p m_p c^2}{r^2} \quad (34)$$

represent electric, magnetic, and gravitational forces. To prepare for its redefinition, a tilde has been added to the gravitational force symbol.

It has been highly advantageous for the electric and magnetic forces to be “commensurate”, as this has enabled the “bending fraction” formalism. For the sun as “circular accelerator”, we now wish to exchange the electric bending fraction η_E with a gravitational bending fraction, η_G . The fact that both electrical and gravitational radial dependencies are “inverse square law”, makes this very natural, but it requires the gravitational law expression to better match the other two laws.

A natural way to do this is to limit the generality of the Newton law to be specific to the sun, which is the only celestial accelerator we are (currently) trying to describe. Fortunately, as accelerator, the sun has a “natural” or “characteristic” bending radius

$$r_0 = R_{A-\text{sun}-\text{lim}} = 6.6 \times 10^9 \text{ m} \quad (35),$$

which is roughly ten times the sun’s radius. We therefore rearrange Eqs. (34) to become

$$F_E = qeE_0, F_M = qe\beta c B_{\text{sun}}(R_{A-\text{sun}-\text{lim}}) \text{ and } F_{G,\text{sun}} = (Gm_{\text{sun}}) \frac{\mathcal{E}_A}{R_{A-\text{sun}-\text{lim}}^2}. \quad (36)$$

In this form the gravitational notation is less than ideal, since, as well as including all nuclear particles, “A” includes all nuclear anti-particles, as well as electrons and positrons. In other words, any non-zero mass, stable, fundamental particles are included. Formally, this gravitational force law excludes massive objects such as meteorites or planets, for which orbits of (small) radius, only 10 times the sun’s radius, are not plausible.

In order to refine the discussion of the electron, proton, positron, anti-positron cosmic ray distributions shown in Figure (5) it is convenient to rearrange the ratio of bending fractions,

$$\frac{\eta_M^*}{\eta_G^*} = q_2 e \beta_2 c B_{\text{sun}}(R_{A-\text{sun}-\text{lim}}) \frac{R_{A-\text{sun}-\text{lim}}^2}{G M_{\text{sun}} \gamma_2(t) m_2} = q_2 \beta_2 \left(\frac{R_{A-\text{sun}-\text{lim}}^2}{G M_{\text{sun}}} \right) \left(\frac{e/m_2}{\gamma_2} \right) c B_{\text{sun}}(R_{A-\text{sun}-\text{lim}}), \quad (37)$$

where the overall sign of the ratio of forces is determined by the initial $q_2 \beta_2$ and final $c B_{\text{sun}}$ products, for which the sign depends on the sign of the sun's magnetic field and on the signs and rotation directions of the four particles under discussion. See the annotations on Figure (5). Notice also that the fractional bending ratio for electrons and protons only become the same for fully relativistic motion, as $\beta_2 \rightarrow 1$. From the figure it is clear that the magnetic and gravitational forces are constructive in all cases. It is (presumably) also true, that the longitudinal Parker electric field causes positive acceleration in all cases.

The only unknown quantity is the Sun's surface equatorial magnetic field B_{sun} , which needs to be established empirically, either by alternate measurement method or by curve fitting based on the present theory. This bending has become negligible in the relativistic limit, but one faces the complication that the electron orbits at low energy depend on radial position $r = R_{\text{sun}} + \delta r$, which is shrinking rapidly in the low energy and semi-relativistic phases. Recall, however, that close to the sun, to first order in $\delta r / R_{\text{sun}}$, the radial magnetic and gravitational force components fall off proportionally. This reduces the sensitivity of the radial magnetic to the gravitational bending ratio and reduces the need for explicit measurement of the radial variation of $c B(r)$.

Complication due to superimposed magnetic bending

What remains to be done is to explain the curves in Figure 5, in order to estimate the actual external gravitational and magnetic fields of the sun by matching the observed centripetal and longitudinal accelerations. Toward this end, we return to the task of refining the gravitational and magnetic bending fractions; GET RID OF gamma2

$$\eta_G^* \approx \left(\frac{G M_{\text{sun}}}{R_{\text{sun}}^2} \right) \gamma_2(t) m_2 \left(1 - 2 \frac{\delta r}{R_{\text{sun}}} \right), \quad (38)$$

$$\eta_M^* \approx q_2 \beta_2 c B_{\text{sun}} R_{\text{sun}} \left(1 - 2 \frac{\delta r}{R_{\text{sun}}} \right), \quad (39)$$

where a reference radius R_{sun} , the radius of the sun, and the magnetic field at the surface is B_{sun} , deviation δr has been introduced and the radial position r is expressed as

$$r = R_{\text{sun}} + \delta r = R_{\text{sun}} \left(1 + \frac{\delta r}{R_{\text{sun}}} \right),$$

where only the leading power has been retained.

Expressed in this way, it can be seen, to a first approximation, that the dependence on radial deviation " δr " of the bending fractions cancels in their ratio. Though the bending fractions still do not sum to 1, their individual meanings are useful approximations, with roughly correct ratio. So, there is some justification in retaining the separate expressions for η_G^* and η_M^* as convenient linearized approximation of the individual bending fractions while treating their ratio as known and fixed.

Apart from the questionable approximations, the only unknown in the bending ratio is the magnetic field, which can be expressed in terms of the sun's surface magnetic field;

$$B_{\text{sun}}(r) = B_{\text{sun}}(R_{\text{sun}}) \frac{R_{\text{sun}}^3}{r^3}, \quad \text{yielding} \quad (40)$$

$$\frac{\eta_M^*}{\eta_G^*}(r) \approx q_2 \beta_2(t) \frac{c B_{\text{sun}}(R_{\text{sun}})}{G M_{\text{sun}} \gamma_2(t) m_2} \frac{R_{\text{sun}}^3}{r^3}, \quad (41)$$

but only briefly, while there is significant bending, say of electrons, resulting from the sun's magnetic dipole moment. The factor $\beta_2(t)$ causes the relative M/G bending ratio to be greater

for electrons than anti-protons while the electrons have become relativistic but the anti-protons have not. However, this effect seems to be more pronounced for the positron/proton ratio than for the electron/anti-proton ratio.

There are two significant time dependencies of $\gamma_2(t)$. One is adiabatic, since its dependence corresponds to the 22 year periodicity of the sun's polarity. It is caused by the variation of the longitudinal Parker electric field component acting on the "temporarily captured" particle being accelerated. Because all three, gravitational, magnetic, and electric need to be accounted for at the same time.

The essential time dependence of γ_2 is the result of monotonic acceleration of temporarily captured fundamental particles. In this case $\gamma_2(t)$ needs to be the solution of Newton's equation for the acceleration caused by the (revised) Newton gravitational force equation. We have to deal with a superposition of three forces, none of which can be neglected, if acceleration of particles of different mass and charge are to be simultaneously accelerated. Since the abscissa in Figure (5) is momentum, it is more careful to describe the functional dependence of γ_2 measurements as $\gamma_2(p_2c)$.

We anticipate finding four solutions, one for each of the four curves plotted in Figure (5). It is essential to realize that the points plotted in Figure (5) are not at all the solutions of Newton's equation; they represent the particle survival fractions, as function's of the originally detected particle types; namely, protons and anti-protons, electrons and anti-electrons. They are experimentally detected particle survival fractions.

Since we are unaware of any theoretical survival calculations for these four processes, it is not easy to explain the low energy curve shapes. But, in a certain sense, the high energy curve shapes explain themselves, in two ways; they are all straight lines on a log-log plot, and, as can be best established with pencil and ruler, the straight lines are all parallel.

Trusting the data plotted in Figure (5), a serious challenge is to account for the ratios of the four exhibited intensities. By first order Taylor expansion about sun's radius, Eq. (39) has been written to linearize the bending fraction deviations in terms of radial deviation δr . Because Figure 5 has log-log dependencies, it seems natural to replace the linear factors in Eq. (39) by negative exponentials.

The analytically evaluated orbit in Figure 15 correlates an energy measured in ISS apparatus apparatus near the earth with a unique orbit that will eventually pass close to the sun. Calling 1 GeV an "intermediate energy proton energy", with relativistic mass greater than rest mass by about $\sqrt{2}$, and extrapolating their subsequent or

Relativistic Kepler orbits

Using special relativity Newton's ellipse planetary formalism is readily modified to respect special relativity.[17]. The leading effect is the in-plane precession of the orientation of the Newtonian elliptical orbit, under appropriate conditions on angular momentum L , is described by the formulas

$$\gamma = \frac{\mathcal{E} + k/r}{mc^2}, \quad (42)$$

and

$$r = \frac{\lambda}{1 + \epsilon \cos(\kappa\theta)}, \quad \lambda = \frac{k}{\mathcal{E}} \left[\left(Lck^2 \right) - 1 \right], \quad \epsilon = \frac{Lc}{k} \sqrt{1 - \left(\frac{mc^2}{\mathcal{E}} \right)^2} \kappa^2 \epsilon \quad (43)$$

This introduction is too abbreviated to follow without referring to the Munoz and Pavic paper. Its purpose here is primarily as excuse to display the orbit shape description introduced initially by Newton:

$$r = \frac{\lambda}{1 + \epsilon \cos(\kappa\theta)}$$

For our conditions, this formula makes understandable a typical relativistic orbit such the one shown in Figure(8). As visualized in this paper, once remote from the sun every cosmic ray particle orbit shape is "geometrically similar" to this one, though not so regular, due to unknown magnetic field perturbations.

Close to the sun, the orbits more nearly resemble the (pseudo)-orbits shown in Figure (15, as continued to enter as well as escaping the sun.

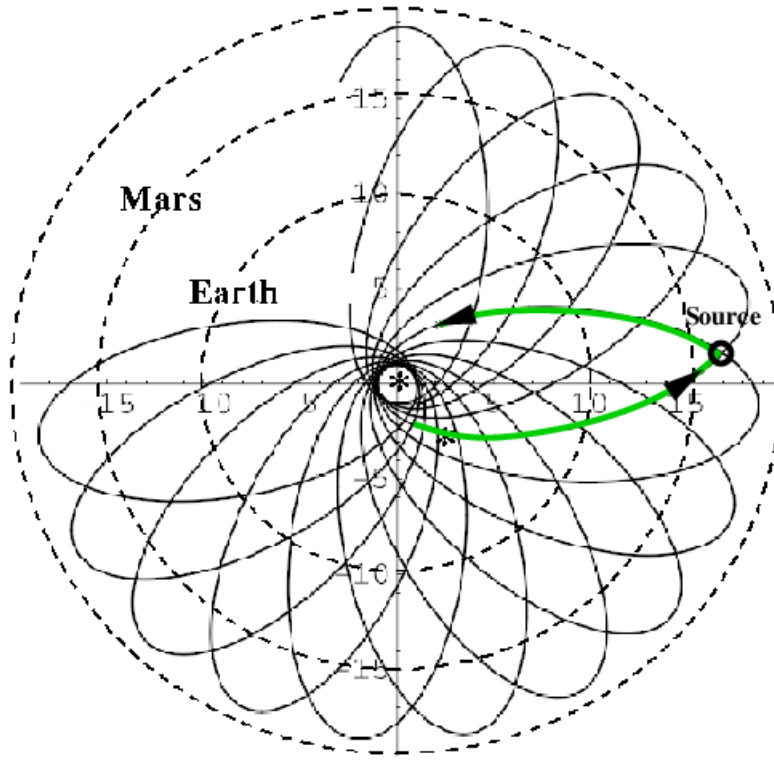


Figure 8: Figure copied from a paper by Munoz and Pavic[17] with a solitary, precessing elliptical orbit representing a single, relatively low energy, cosmic ray orbit. This particle is intended to stand in for an entire atmosphere of cosmic rays. In such a distribution there would be balanced in-flow and out-flow through spherical surfaces centered on the sun. The red arc beginning at “Source” represents a “balancing orbit” passing down through the earth’s orbit.

Voyage to the sun and back

Figure 9 is related to the red arcs in Figure 8. It represents a thought experiment employing a FNAL Tevatron accelerator situated on Mars and aimed toward the sun. This is not unlike the actual FNAL experiment, with Tevatron neutrinos *bent down by 7.5 degrees*, and aimed at an underground detector site in South Dakota.

To match the red arcs in Figure 8 the Tevatron would have to be situated on Mars, at the point labeled “Source” (with the beam *bent up by, say, 5 degrees*, to reduce beam damage caused by interactions with the Mars atmosphere). Comparing Figure 9 with Figure 8, one sees that the Tevatron is not pointed correctly, It would need to be pointed significantly to the right, in order for the return beam to return to the FNAL site on Mars. Also, beam steering caused by the sun’s magnetic field would need to be taken into account.

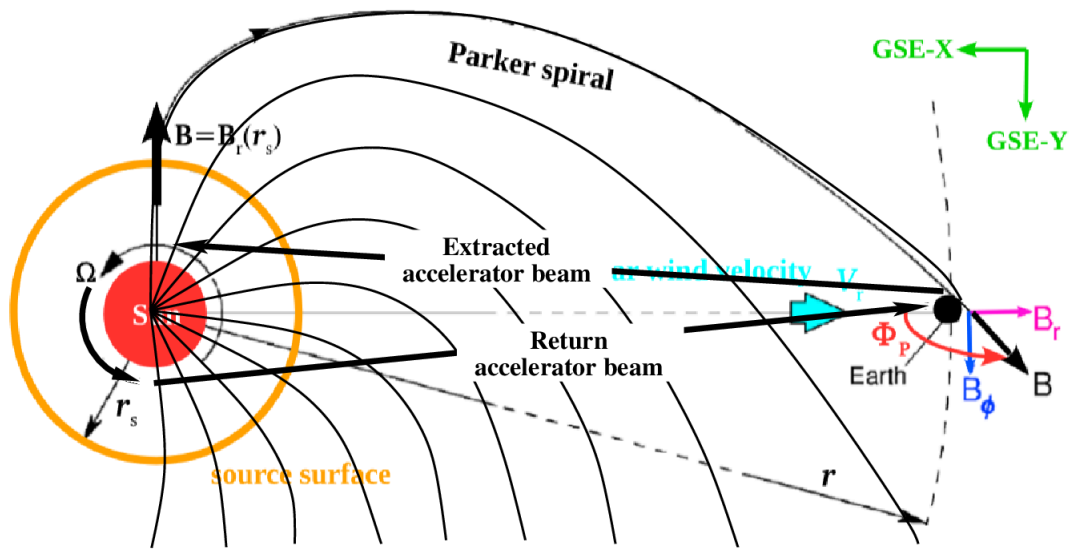


Figure 9: Experimental “thought apparatus” sending a Mars-based Tevatron beam on a “voyage to the sun and back”.

“Improving” the performance of solar particle acceleration

Continuing our effort to design, or rather to understand the design, of cosmic accelerators, an essential ingredient is the longitudinal electric field component $E_{\parallel}$ that is causing the beam particle energy $\mathcal{E}_2(t)$ to increase with time;

$$\mathcal{E}_2(t) = \mathcal{E}_{2\text{inj}} + q_2 E_{\parallel} t. \quad (44)$$

At this point we make a few significant observations:

1. We have been assuming the Parker longitudinal electric field component $E_{\parallel}$ is constant, causing the beam energy to increase monotonically. This would seem to violate the E&M requirement that the line integral of static longitudinal electric field must vanish.
2. The task seems, therefore, to be impossible. But wait; the Sun’s electric field is not constant; it oscillates with a period of 22 years. Perhaps, or even probably, this adequately overcomes the item 1 constraint. In any case, the direction of stable particle rotation around the sun certainly must reverse every 11 years. During reversals, when the magnetic field vanishes temporarily, the launching of cosmic rays into space must stop, at least temporarily.
3. It is impractical, however, to suppose that the entire process of populating the solar system with cosmic rays needs to be repeated every 11 years. There are too many empirical observations that would refute this possibility. It must, therefore, be the case that the vast majority of cosmic rays represent a nearly-constant-in-time equilibrium in which the replenishment of the equilibrium is briefly interrupted ever 11 years.
4. We are striving to accelerate a beam, say a proton beam, from an energy, of say 10^{-2} GeV, to an energy of, say, 10^6 GeV, which is just slightly higher than the highest energy cosmic ray proton energy plotted in Figure (11). This is eight orders of magnitude acceleration.
5. Previously, in Section (2), based on the known magnetic field at the surface of the sun it was estimated that protons could be accelerated from 2 MeV to an energy of 200 GeV in a single revolution, an acceleration by a factor of five orders of magnitude. Though impressive, this is not even close to eight orders of magnitude. We are still missing three orders of magnitude.
6. The whole point of a ground based particle accelerator is to increase the energy a little bit each turn, for many turns. All we need is for the cosmic ray beam to stay captured by the sun for a thousand optimal turns or, at least, to make many thousands of less than optimal turns around the sun. This would seem to be easy, considering that beams traveling at (almost) the speed of light have been stored in ground based accelerators, with tiny apertures, for day-long runs.
7. In actuality, in real life, the situation is more complicated than has so far been assumed. However, we need not require 100 percent beam capture, nor perfect beam flux retention through the acceleration process. According to Figure (12), the extracted beam flux at maximum energy is less by 12 orders of magnitude at maximum compared to minimum energy.

6 “Double slingshot” (failed) solar cosmic ray mechanism

This section proposes (but rejects, as unworkable) a “double slingshot” mechanism for generating cosmic rays within the solar system. The rejection is based on the absence of plausible injection and extraction mechanisms resembling laboratory-based particle acceleration. This material is retained, for now, to illustrate the futility of treating cosmic ray production with

mechanisms too closely similar to laboratory accelerators, and as reminder of the time and space scales involved.

To begin, the sun’s extracted solar wind serves as multi-particle injector, onto Jupiter as pre-accelerator, of nuclear isotopes onto orbits in the equatorial plane above the equator. Only orbits on the side of Jupiter for which the force is centripetal will be (temporarily) captured. Here the magnetic field is parallel to Jupiter’s axis of rotation, with a magnitude of approximately 4.2 Gauss at the surface. Other rough parameter values have been provided in the previous Section 2.

The shape of Jupiter’s magnetic field is essentially the same as that of the magnetic field of the sun, which is shown in Figure 2. After one or more partially complete helical turns around Jupiter, the much accelerated isotopes will have extracted themselves into a fan of energetic nuclear isotopes emerging isotropically in the two dimensional equatorial plane external to Jupiter. This is illustrated in Figure 10.

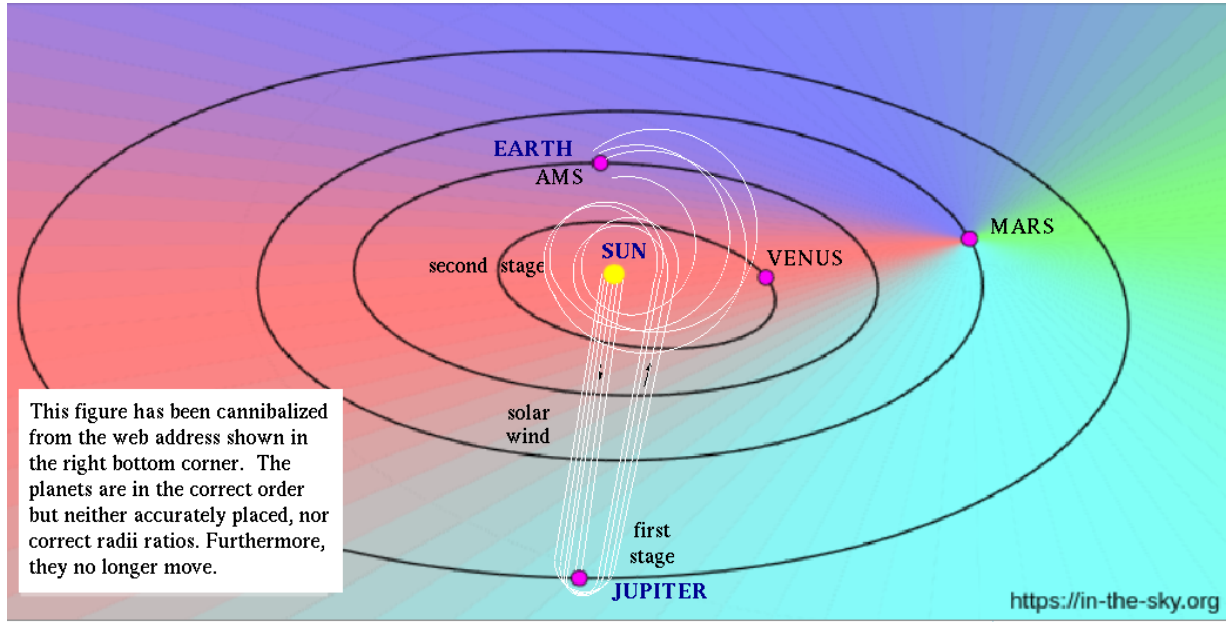


Figure 10: Artist’s conception of two stage “Double-Slingshot” solar system production of cosmic rays, impinging, for example, upon the Alpha Magnetic Spectrometer (AMS) detector[3] aboard the International Space Station (ISS). As indicated in the corner note, planetary distances from the sun are much distorted. Mainly, the radius of Jupiter has been reduced, relative to the other planets, including earth, by roughly a factor of two. Slingshots first around Jupiter, then around the sun, produce acceleration factors discussed in Section (2). Of the six orbits shown leaving the sun towards Jupiter, two could potentially be detected near Earth by the AMS detector.

It is pictorially attractive to visualize a “two-stage” celestial copy as a laboratory-based accelerator complex, such as shown in Figure 2. However, neither injection into nor extraction from the sun (as storage ring) has been proposed. This figure is retained for reference, should such mechanisms be proposed. It is also useful for exhibiting the time scales necessarily present in solar cosmic ray production, as well as further understanding features of the sun as accelerator.

Even the least energetic of these isotopes aimed exactly toward the sun will arrive there in less than a day, where they are injected into the sun’s exterior magnetic field, which is playing its role as a circular accelerator. Half of these injected nuclei will be deflected away from the sun, back into space. The other half will be attracted toward the sun. These will mainly crash back into the sun. But a significantly large number will have low enough momentum to commence slowly expanding or contracting helical motion around the sun.

In this situation, these orbiting particles (probably predominantly protons, but some doubly energetic deuterons, and quadratically energetic α -particles, and some even more energetic

nuclear isotopes) will continue cycling helically around the sun. In this situation there is no way of telling (short of a careful modeling of the overall process) the maximum energy these particles can obtain. As a guess, one hopes/expects the energy distributions will match those of various the measured distributions shown in Figure 12. It is not ruled out for the occasional nuclear isotope to make more than one passage through one or the other (or both together) of the slingshots. At least in principle this could account for even the highest energies shown in Figure 12.

7 International Space Station (ISS) cosmic ray results

Properties of the Alpha Magnetic Spectrometer (AMS-02)

The quality of the cosmic ray data from the ISS over the last twenty years is outstanding; a treat to pore over and understand. The data in Figure (5) alone provides enough data to confirm or refute a significant fraction of the cosmic ray theory presented in this paper. But the critical experimental data most needed to test my solar cosmic ray picture is quite weak at present. Rather than scanning all cosmic ray experimental data, this section explains why the AMS magnetic spectrometer data is blind to the data that would provide the most critical test of the solar cosmic ray picture.

By design, the AMS-02 spectrometer is optimized to provide universal coverage of cosmic rays aimed more or less in the direction pointing toward the sun from the location of the ISS which, for this purpose, is the same as the location of the earth. The acceptance cone of the AMS, though large compared to laboratory spectrometers, is conical, with conical acceptance angle of approximately 0.5 steradians, which is quite small compared to the 4π needed for full angular coverage.

Measured in space within the solar system, incident cosmic ray directions are more or less isotropic, with differential energy distribution falling with power law, $\mathcal{E}^\gamma$ spectrum, with “spectral index” $\gamma \approx 3.0$. The several plots in this paper of this energy distribution, as measured in the ISS, agree well with this behavior, modulo small “kinks”, as has been described in detail above.

My solar cosmic ray picture is consistent with this isotropic, power law energy distribution. In this picture, should the AMS-02 apparatus have been pointed away from, rather than towards, the sun, it would be observing much the same cosmic ray distributions as at present. Unfortunately (in this respect) the AMS-02 spectrometer was not designed for isotropic coverage. In fact, the 17 radiation lengths thick electromagnetic calorimeter (ECAL), present to measure the energy released by electromagnetic particles, terminates the AMS-02 beam line. Along with the electronic vetoing of sideways traveling cosmic rays, the AMS-02 apparatus is, by design, insensitive to the angular distribution prediction of the present paper.

An apparatus “perfect” for testing solar cosmic ray generation, would have a pair of AMS-02 spectrometers, pointed in opposite directions, one towards, the other away from the sun.

ISS-measured isotope cosmic ray survival vs energy

Many cosmic ray detection experiments have been performed aboard the International Space Station (ISS): PAMELA, AMS02, ATIC, CREAM, CALET, DAMPE, NUCLEON, etc. For example, the AMS02 experiment is responsible for the splendid data in Figure 11. Energy spectrum measurements from some or all of these experiments are shown in Figure 12, in the form of a lower energy data set in the bottom figure, as well as a combination of the same lower energy data and a higher energy data set in the top figure.

Nearly perfect negative exponential energy dependence is shown in these plots. Because these are log-log plots, the pure exponential dependence can be quantified in simple decibels per decade terms, as indicated by the inset calculations. Though not calculated very carefully, the slopes are sufficiently equal to suggest that all these data sets have the same origin, either

solar or galactic, but certainly not both. But, looking more carefully, there seem to be “kinks” in the plots of individual nuclear isotopes.

Since much has been made of the proton energy dependence in this region, this is pursued in the subsequent figures and discussion, centering on Figure 13, which is copied from Lipari and Vernetto reference [15], who performed multi-parameter curve fitting algorithms devoted entirely to analyzing the energy region from 10^2 to 10^5 GeV. In order to enhance variation through this region they factor out the average slope to amplify the “kink” region visually.

In one sense the Lipari-Vernetto fitting procedure was a good idea. It shows that all seven experiments, presumably analyzed independently, agree quantitatively as to the existence and magnitude of the kink effect under study. On the other hand, the expanded energy dependence, is highly misleading. Sorting this out is, however, hugely instructive, and is the subject of much of the following discussion.

The analytic representation of the exponential decay of voltage as a function of time in an RC circuit is elementary. It is the unambiguous solution of a linear differential equation. It meets the requirements required to be a Markov process, in that the past is constantly being forgotten. The decay of detection probability as a function of energy, though less elementary, seems to be essentially similar, with the exception of an occasional kink. A simple physical interpretation for this kink is that there is a sudden discontinuity in the proton energy. Once again, this could not be said to be “predictive”. It could only be referred as “descriptive”.

In the days of the “Star Wars” movie, the clever description of such a phenomenon, coined by Etienne Forest, was “beam me up, Scottie” propagation, which is, of course, impossible. What is possible is that, over time, the energy changes but the position does not. This phenomenon is most commonly observed in circular accelerators as the beam passes through an RF cavity, treated as having zero length. This is a “sudden” approximation, of the sort that is commonly quite accurate.

As an experimental interpretation, this “proves” that a significantly large fraction of the protons being detected have been accelerated, in principle with either sign, though in practice always positive. as they stay in contact with the sun. This can only have happened during temporary circulation around the sun. Once escaping the sun, the particle survival has deviated slightly from its previous value, and causing a small kink in the data.

The data plotted in Figure 13 is a spectral distribution, energy spectrum in this case, for which “local curve fitting” is not at all appropriate. Circuit currents “decay”, but they never “un-decay”. But particle energies can change and their apparent survivability must change correspondingly.

As with the Heisenberg uncertainty principle, the effect of an instantaneous impulse is spread over a large range of energies. What makes the kink in the proton energy distribution most persuasive is that all seven experiments noticed the same impulse. Since the protons are fully relativistic, their time and their orbit radius known, the fractional revolution around the sun can be determined. Then, by their energy change, which seems to have been, by a factor of one hundred, according to the kink, or by a factor of 10^5 in Figure 14. In passing, one notes that the longitudinal Parker electric field component could therefore be said to have been measured approximately.

ISS measured cosmic ray fluxes

The closely matching “shapes” of ISS plots argues that all the cosmic ray data is either solar or galactic, but not both. There has, however, been a curious kink in the cosmic ray energy distribution, observed persuasively by seven independent experiments on the ISS have been subject to the same impulse.

There is a further complication, which will be ignored for the time being. Along with the gravitational bending, there is also magnetic bending. Though the acceleration is caused only by the Parker electric field, the centripetal bending, which is predominantly gravitational, is partly magnetic. The major theoretical advantage from ignoring magnetic bending is that orbits subject only to gravitational bending are invariant under time reversal.

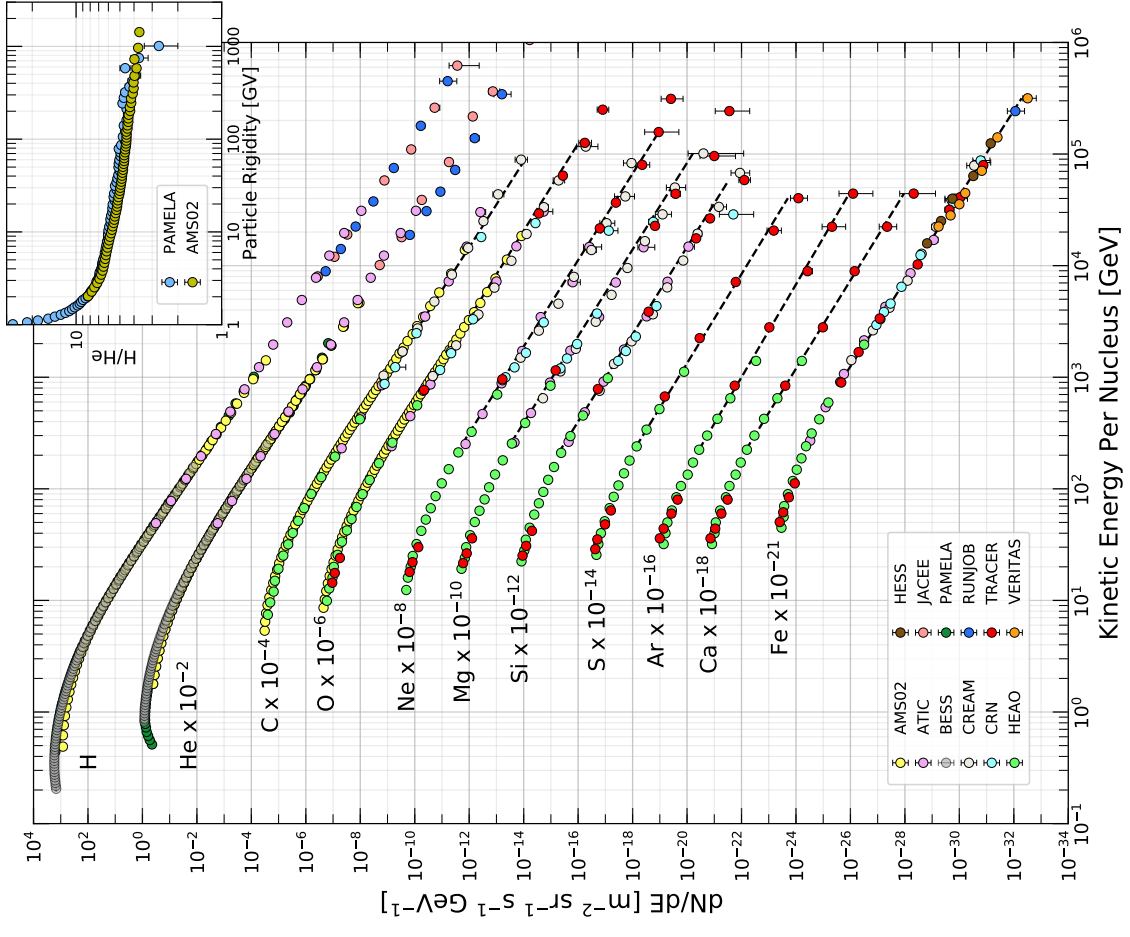


Figure 11: Figure copied from Particle Data Group, Chap-29, Cosmic-Rays. showing energy dependence and relative abundance ratios measured by AMS within the ISS. This plot is superficially misleading by suggesting it is representative of all atomic nuclei. In fact, the “light elements” (Li, B), and especially Be, abundances are not shown; presumably because these nuclei are so anomalously missing as cosmic rays.

The helical orbits in Figure 15 are measured by AMS to have the same energy, augmenting by one, the number shown of protons indicated as already present in the bin labeled 500. There are numerous points to make about orbit tracking in the AMS detector, and related to Figures 15. and Figure 16. Let us say they correspond to one of the six events labeled in Figure 16 To understand the figures it is essential to have available the original paper [16], while assessing the following points:

1. The tracking measurement is made near the earth, while the orbits are accurately reconstructed near the sun. No matter how carefully the orbit has been aimed, it is clear that, during the fitting process, the orbits have been constrained near the sun, based on the measured proton energy.
2. Each orbit does not represent a single proton, nor even a single beam of protons. It is an orbit constructed from points detected in a particular energy range over many years, and binned accordingly. Clearly the jagged nature of the orbit reflects fluctuation effects that are prominent at low proton energy, but which become less pronounced with increasing energy (as expected from unknown magnetic fields encountered) Already by 1 GeV, with proton velocity $\beta_p = 0.87$, the orbits are nearly perfect circles or seemingly, shrinking helices.
3. One can see from these figures how measured flux per bin as ordinates (in figures like Figure 12) can be correlated with matching bin energy as abscissas.

4. Though these figures are naturally analyzed as helices shrinking as they approach the sun, it may be possible that some of the points are arriving at the sun, and some are leaving the sun.
5. This confuses the interpretation of these reconstructed particle orbits as representing an equilibrium distribution of more or less isotropic protons and other cosmic ray particles.
6. In equilibrium there need to be approximately the same number of cosmic rays going either way, to or from the sun. However the AMS detector acceptance is limited to a narrow solid angle of directions pointing toward the sun.
7. Forward and backward orbits are spatially identical, but their time evolutions are opposite.

8 Markov model of cosmic rays as accelerator beam particles

Measured solar cosmic ray distributions

Continuing the interpretation of proton cosmic ray data, Figure 16 is a *tour de force* plot of proton flux density measurements, measured continuously from 2011 until 2019 (and to this day, but data not yet published). There is also ISS data for all nuclear isotopes. It is displayed in Figure (11). To display such a vast amount of data, they bin the data day by day, producing and plotting a new point every day—eventually for the 22 year period of solar magnetic field oscillation, but, for now, just for 10 years.

The ISS is situated close to the earth. For the comfort of the astronauts, the orientation of the ISS cannot be rotating very rapidly, and it must also remain more or less upright, with axis pointing toward or away from the sun. This is important also for the cosmic ray detection apparatus, which provides precision particle tracking, primarily over the relatively small solid angle, as imposed by the limited detector mass.

Though the cosmic rays are more or less isotropic, the detection apparatus is oriented to emphasize incoming cosmic rays aimed more or less toward the sun. Fixed relative to the spacecraft, it is quite challenging to fix the cosmic rays relative to, say, a coordinate system with origin at the center of the sun, and orientation fixed relative to the fixed stars. It is likely true, also, that most of the ISS experiments cannot distinguish whether events are headed towards or away from the sun. Even the AWS detector is short enough to make it challenging to establish the sign difference between “entrance” and “exit” times.

There is a procedure in differential geometry, using “Cartan moving frame” formalism, that accomplishes this task. With momenta inferred from energy, and presuming Newton’s gravitational law to be valid, a single point on the cosmic orbit can be established remotely, from an observation point near the earth. Of the six orbits established (and labeled at the bottom of the upper plot, in Figure 16. the highest momentum “proton” follows a circular orbit closest to the sun, captured at least temporarily by the gravitational attraction toward the sun. As can be seen by the labeling of the figure, what is plotted along the radial axis, is the measured proton flux through the magnetic spectrometer in the ISS. As already stated, this is all very challenging. Yet there is a further challenge.

Even though the Cartan reference frames have been established, there is a time delay issue, resulting from the time delay equal to the transit time of the protons from the ISS to the sun, which is a distance of approximately 1 AU, or 1.5×10^{11} m. Traveling at the speed of light the time delay is about eight minutes. For a 1 GeV proton, γ is approximately 2.5, and the speed is about 0.87 c.

This is a large enough difference to account for the ragged 1 GeV proton “orbit” shown in Figure, 16. It can be seen, however, in the lower figure, that the 10 GeV orbit around the sun is very nearly closed. Note also, that this same orbit matches almost exactly the white circular orbit, labeled “ r_2 limit circle”, which is the circle that fully relativistic charged particles,

As already mentioned, the Cartan orbit reconstruction is time independent. If an orbit respects CPT invariance, the Cartan tracking cannot distinguish between forward and backward track following, especially if magnetic bending is weak.

With these considerations, it seems plausible to suppose that the orange curve, defined by ragged data points, can be interpreted as either the entering to, or the exiting from, curve of the 1 GeV protons.

Original and acceleration-imposed Schottky noise

There has always been an intimate connection of statistical physics with accelerator physics. Particle beams of a single species have statistically distributed distributions in six dimensional phase space. Though these distributions limited the performance of early (linear) accelerators, the statistics was quite simple because the only one way of improving beam precision was by collimation, necessarily reducing the counting rates.

The advent of circular accelerators changed this situation dramatically. With long term stability understood, and good vacuum achieved, the same beam could impinge repeatedly on the same target, hour after hour. Though eliminating the previous problem, this introduced a new statistical problem, namely beam quality reduction caused by the stochastic scattering of every beam particle in the target, causing stochastically increased “emittance growth” and corresponding loss of particle phase space density, *which would violate Liouville’s theorem*. This, too, was reasonably easy to understand, but could only be reduced by reduced target thickness and counting rate. Unlike the original beam spreads this spread is “induced” by the beam acceleration mechanism.

In the sun as beam accelerator, both of these noise sources are present and both can, in principle, be measured. This is why Markov chain treatment is appropriate for diagnosis of the sun as accelerator. Markov processes are processes with “no after effect” meaning the past is always forgotten; the future is based only on present conditions and future sources of noise.

In the present context, all of this discussion could not be said to be “predictive”. It could only be referred as “descriptive”, based on the experimental data. Most of the following discussion is subject to the same qualification.

A more serious application of stochastic processes in accelerator physics came with the introduction of detectors sensitive to “Schottky noise”, both “longitudinal” caused by arrival time spread, and “transverse” caused by coupling of longitudinal and transverse coordinates. Then came stochastic cooling, which comes as close as possible to serving as a “Maxwell demon” violating the second law of thermodynamics. For the sun as accelerator, *neither of these topics is applicable*, since the observer has no control over the external signals influencing the statistical properties of the beams being accelerated.

“Liouvillean” cosmic ray high energy flux dependence on energy

Figures such as Figure (5) and Figure (11) provide information concerning the energy dependence of the flux of various particle types.¹¹ From these plots one can infer the energy dependence of abundance ratios of pairs of particles. Superficially the abundance ratios themselves show no strong energy dependence. (i.e. the “shapes” of the curves are similar). Especially striking is the fact that, once relativistic, these ratios seem to be “frozen”, independent of energy as shown by the identical high energy slopes, independent of particle type.

At the same time, a figure such as Figure (11) shows abundance ratios varying over 20 orders of magnitude, all with the same flux vs energy dependence, at fully relativistic energy. Simple description of this behavior is that, for fully relativistic particles, the acceleration mechanism is “the same” in all cases, independent of charged particle type. Once again, this could not be said to be “predictive”. It could only be referred as “descriptive”.

¹¹Technically “rigidity” is “momentum per charge”, not “energy”, but, at fully-relativistic energy this distinction has become inessential.

A universal feature of radiation from sources of small size, is that the intensity (expressed as the flux in this case) varies inversely as the square of distance from the source. This represents conservation of energy. Plotted on log-log paper the slope would be two decades of flux for every one decade of energy.¹² However, Figure (11) represents the distributions in energy, not radius. The data itself, for example Figure (5), shows 3.3 decades of flux per decade of “rigidity”, meaning “momentum per charge”, with momentum and energy. It is this data that converges to parallel straight lines at high energy.

We are familiar with straight lines on log-log plots representing the semi-permanent survival of long-lived, but unstable, nuclear isotopes. Starting with equal numbers of a stable and an unstable isotope, the plot of their surviving fractions would be identical only until the last of the unstable particle had decayed.

One can account for this by noting that only stable nuclei are displayed and unstable nuclei are both rare and not plotted. We know, for example, that there are few enough Be nuclei to justify their not being included.

Something still seems surprising about Figure (11). The data seems to show in some mechanical sense, that all charged elementary particles are “identical”. Looking closely at the straight line fits to C, O, Ne, Mg, Si, S and Ar, it can be seen, in almost every case, that there are actually kinks in the data which seem to be to be matched by two, rather than one, broken fit line. This is consistent with the requirement, following the curves from right to left, for them to roll over to be more or less horizontal at the left edge of the graph. But why should the asymptotic extrapolations be exactly parallel in the fully relativistic limit? Perhaps this was imposed by the choice of energy scaling factors?

The absence of lithium, and (especially) beryllium, and boron, has always been ascribed to the ultra-short 10^{-18} s lifetime of $^8\text{Be}_4$; its absence from the solar wind is discussed in reference [1].

This provides a warning to be careful about possible mis-interpreting of the data. Figure (17), copied from Roman and Bartoletti, [10], “A master equation for power laws”, illustrates a situation much like the present one. This is explained in the caption to that figure. It is also discussed in the sequel.

In our model of the sun as accelerator there are periodic sources of “noise”, encountered while individual particles pass by the sun. In the high energy limit, where all particle energy fits are parallel, the same stochastic fit applies to every particle type, but the incoherence of the perturbation requires the iterated fit to more nearly fit the measured data, in spite of the quite large energy range of the individual fits.

Systematic increase in solar cosmic ray particle energies

A valiant effort has been made so far, based on “established physics”, meaning, primarily, special relativity, to account for observed cosmic ray properties. This section contradicts nothing described so far. Rather, it reformulates the astrophysics of cosmic rays starting from a Hamilton-Jacobi (H-J) wave-like interpretation of Newtonian gravity. This is in contrast with Einstein’s general relativistic formulation of cosmology.

The Jacobi partial differential equation reformulation of Hamilton’s classical mechanics is reviewed in Appendix A, *Hamilton-Jacobi Wave Mechanics*, which has been generalized to account for special relativity, and patterned after Schrödinger wave mechanics, but not encumbered by quantum mechanical complications, such as failure of commutation and Heisenberg uncertainty, to be sufficient for understanding cosmic ray generation.

There is a solitary experimental cosmic ray feature that complicates our understanding of cosmic rays. It was hinted at in the previous “Disambiguation of the term ‘cosmic ray’ subsection” of this paper. Now it is returned to in Figure 8 which exhibits a class of particle orbits in an appreciably large central portion of the solar system. The figure also serves as a reminder

¹²Especially in accelerator x-ray sources, this radiation property is expressed as “brightness” or as “brilliance”, two quantities that are technically different but, for present purposes, are much the same [11].

that, from the earth’s perspective, the solar and galactic systems are both “astronomically large”.

Just a single celestial orbit is plotted in Figure 8, but in its various phases and amplitudes it can represent a multiplicity of different particle orbits constituting the distribution of “light” (meaning atomic and nuclear) material in the solar system. An important observation is that there are as many particles aimed away-from, as aimed towards the earth. This satisfies a major requirement of equilibrium in the cosmic ray atmosphere. Plotted in 1950, Figure 8 takes proper account of special relativity to explain the dependence of particle velocity on distance from the sun. Plotted here it does not yet account for the stochastic, more or less monotonic, increase of single cosmic ray particle energies during their brief but repetitive circuits around the sun.

Hamilton-Jacobi description of accelerating particle orbits

To begin with, it is important to qualify the meaning of “unperturbed” in this section. The basic understanding of terrestrial particle accelerators is said to be “exact”, even though engineering defects, and only partially understood “second order” effects persist, that cause “unperturbed” to be somewhat of an exaggeration.

For our solar accelerator, the meaning of the term “unperturbed” needs to be qualified far further, even to the point of making the term misleading. Nothing concerning injection can be said to be “exact” even for laboratory-based accelerators. However, the term applies quite nicely to the subsequent acceleration and extraction processes. Anything said about injection into or extraction from, the solar accelerator, can only be conjectural.

For the sun as accelerator, the acceleration phase is described by closed-form Hamilton-Jacobi formalism that provides unambiguous answers that can be correct or incorrect, depending on the truth of the assumptions and the accuracy of the parameters. This is the only sense intended as the meaning of “unperturbed” in this section.

Our model of solar cosmic rays is based on charged particles following relativistic Keplerian orbits. Not yet described, however, is a quantitative calculation of the gradual acceleration of particles during the brief sections of their orbits as they temporarily circulate around the sun. For this calculation, a Hamilton-Jacobi partial differential equation description of the relativistic mechanics seems to be appropriate. The special relativistic Kepler orbit treatment is already “time-dependent” by virtue of the time dependence implicit in $\gamma_2(t)$ variation; we are now introducing the further time dependence associate with acceleration by the sun’s longitudinal electric field.

Formally, this new time dependence could simply be accounted for by a more detailed evaluation of $\gamma_2(t)$. But this is easier said than done. A perturbative treatment seems needed in order to calculate the further dependence on time of the orbits during the brief time they are circulating around the sun.

This (more or less periodic) time dependent acceleration is different for different particle types. It can best be treated as the perturbation of otherwise energy-conserving relativistic Keplerian orbits. The unperturbed H-J description of these orbits is described in Appendix A, *Hamilton-Jacobi Wave Mechanics*.

Calculations like this have become routine in Celestial Mechanics (again with the simple generalization, in our case, from non-relativistic to special relativistic mechanics). In fact, the most general treatment is due to Lagrange himself, in his “Lagrange Planetary Equations”, which introduced adiabatic invariant “Lagrange brackets”, which were later inverted into “Poisson brackets”, and later yet, by Dirac, into quantum commutation relations. Much of this is described in references[12] and [9].

9 Proposed test of this solar cosmic ray source model

Many aspects of solar cosmic ray astrophysics have been addressed in this paper. All have been based on the proposition that rest-masses need to be replaced by relativistic energies as the inertial factors in Newton’s inverse square law radial force dependence. Many consequences

of this alteration have been described, all of which seem to be consistent with present day observations.

At most one, but not the other, of the extra-galactic or the solar source models can be correct. Based purely on presently available experimental data, one cannot decide which is more nearly correct.

An experimental test is proposed which may be workable and might be expected to favor one or the other of the purely solar or the extra-galactic cosmic ray source possibilities. This proposal places two identical but anti-parallel AMS spectrometers on a next generation ISS space ship.¹³ Comparison of the symmetries of the cosmic ray distributions measured by these two detectors should help to distinguish between solar-based and extra-galactic-based cosmic ray models.

For the extra-galactic model, perfect initial cosmic ray isotropy is assumed, and currently observed asymmetry is ascribed to shielding by magnetic fields encountered by cosmic rays after their entry into the solar system. For the solar-origin cosmic ray model, there will also be observed asymmetry. But the details of these two cosmic ray anisotropies will be very different.

What has not yet been provided in this paper, is a self-consistent modeling of existing cosmic ray data, based on formulae derived in this paper. Ingredients for such a model have been described, including the H-J description of individual particle orbits, dominated by their brief but repetitive encounters with the sun. Concerning these orbits while remote from the sun, the absence of coupling between particle energy and the other three H-J separation momenta helps reduce uncertainty associated with orbit wandering caused by unknown large scale magnetic forces.

Comprehensively measured abundance-ratios have been displayed but not yet carefully accounted for using theoretical methods described in the paper. Some of these comparisons should be quit easy; especially $p/p\text{-bar}$ and e^+/e^- energy-dependent ratios, which are especially challenging for extra-galactic sources. The anomalously low values of Li, Be, and B, light-element isotopes, with their solar-wind sources, as usual, need to be explained.

¹³This is not as easy as it sounds, since the solar panels for the ISS, always shield the ISS from the sun, possibly impeding detection of cosmic rays coming directly from the sun.

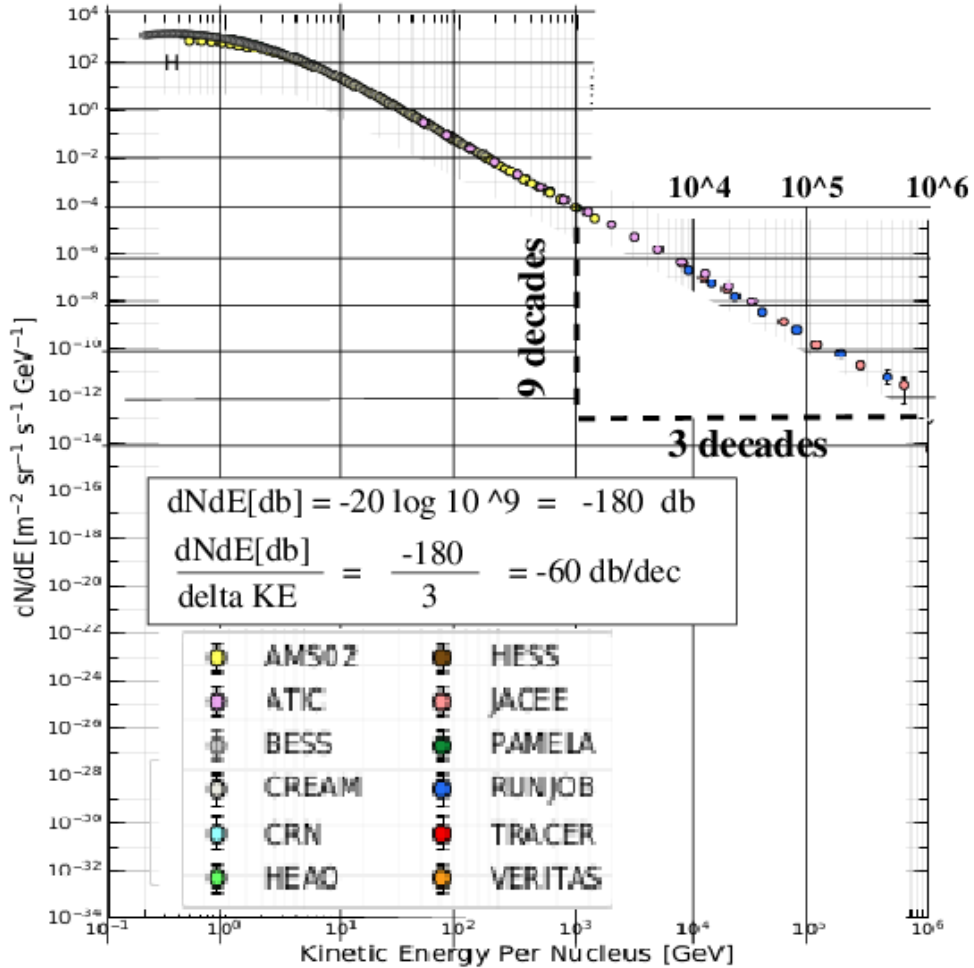
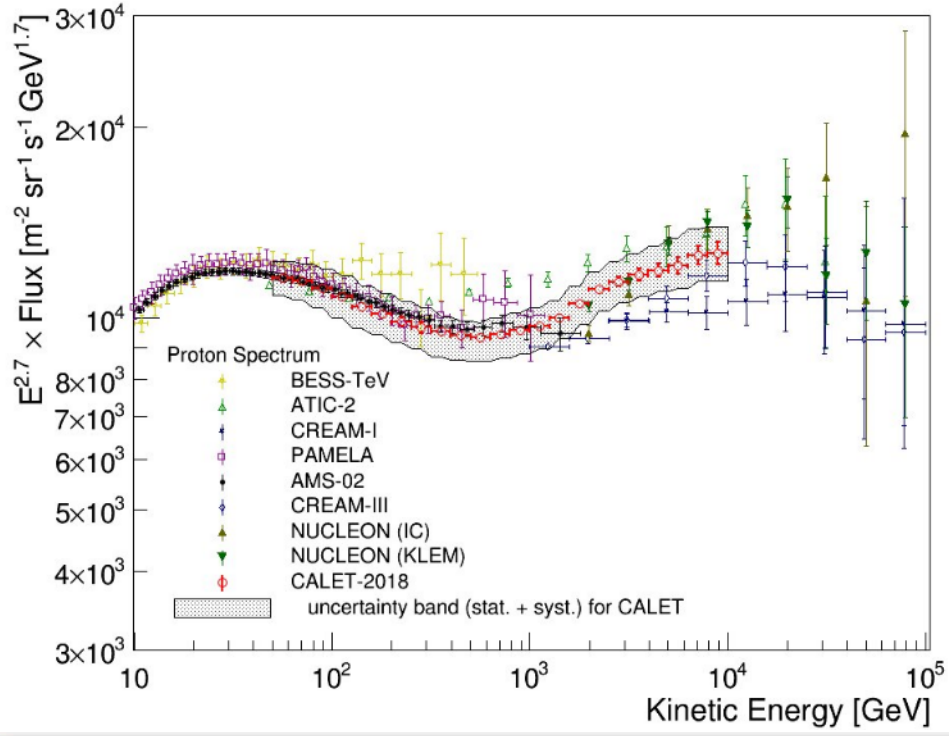


Figure 12: ISS measured flux of cosmic ray protons, plotted as function of proton energy: **Top:** dN/dE flux dependence, with best fit exponential factor $KE^{2.7}$ scaling; **Bottom:** unscaled data, with best fit exponential decay (expressed as db/decade). This slope of (approximately) 3 decades per decade only matches the fully relativistic data, which is more or less consistent with the $E^{2.7}$ energy scaling in the upper figure. This proton data is the same as the more complete data shown in Figure (11).

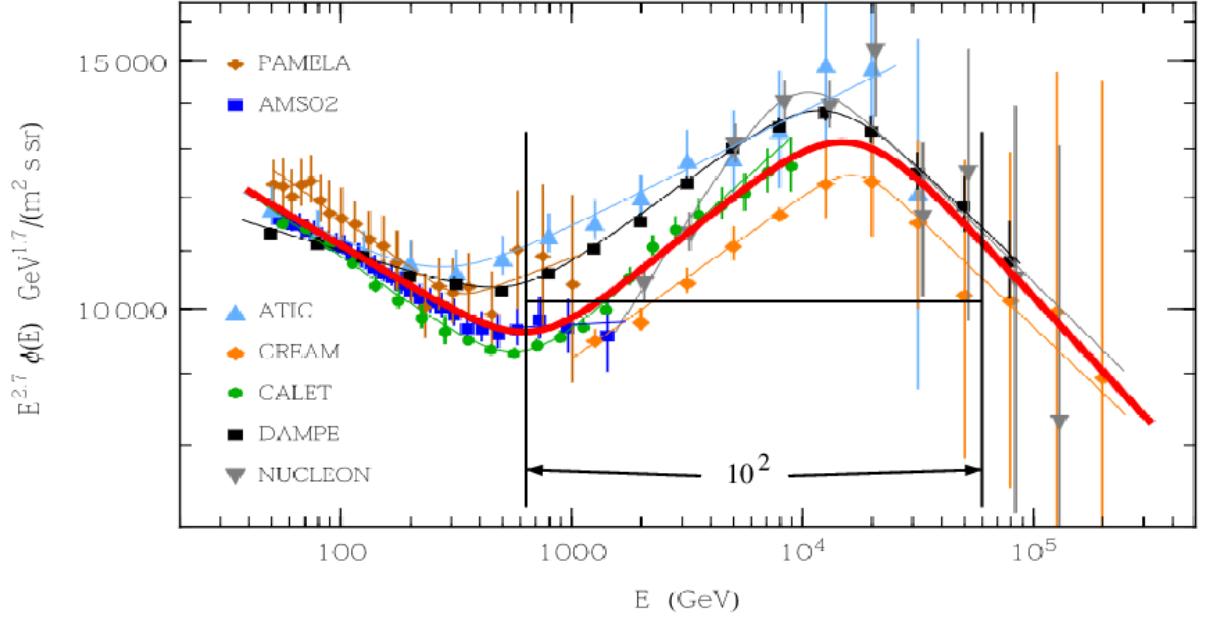


Figure 13: This figure has been copied from Lipari and Vernetto[15]. This data shows that all seven experiments on the ISS were subjected to the same impulse, or that all protons were subjected to the same impulse when near the sun. In the present paper this is interpreted as acceleration of protons from 600 GeV, to 60,000 GeV, while briefly circulating around the sun.

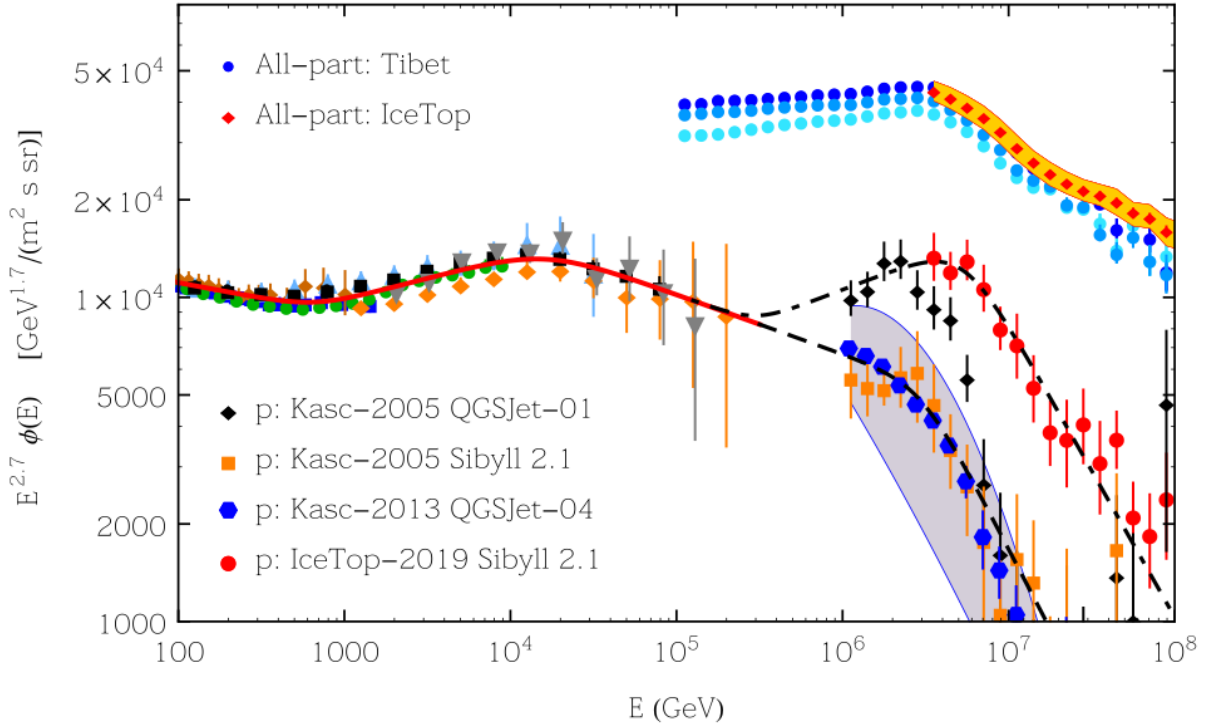


Figure 14: Also copied from Lipari and Vernetto[15], the data in this plot, increase the acceleration range, starting from 100 GeV and running almost to 10^7 GeV.

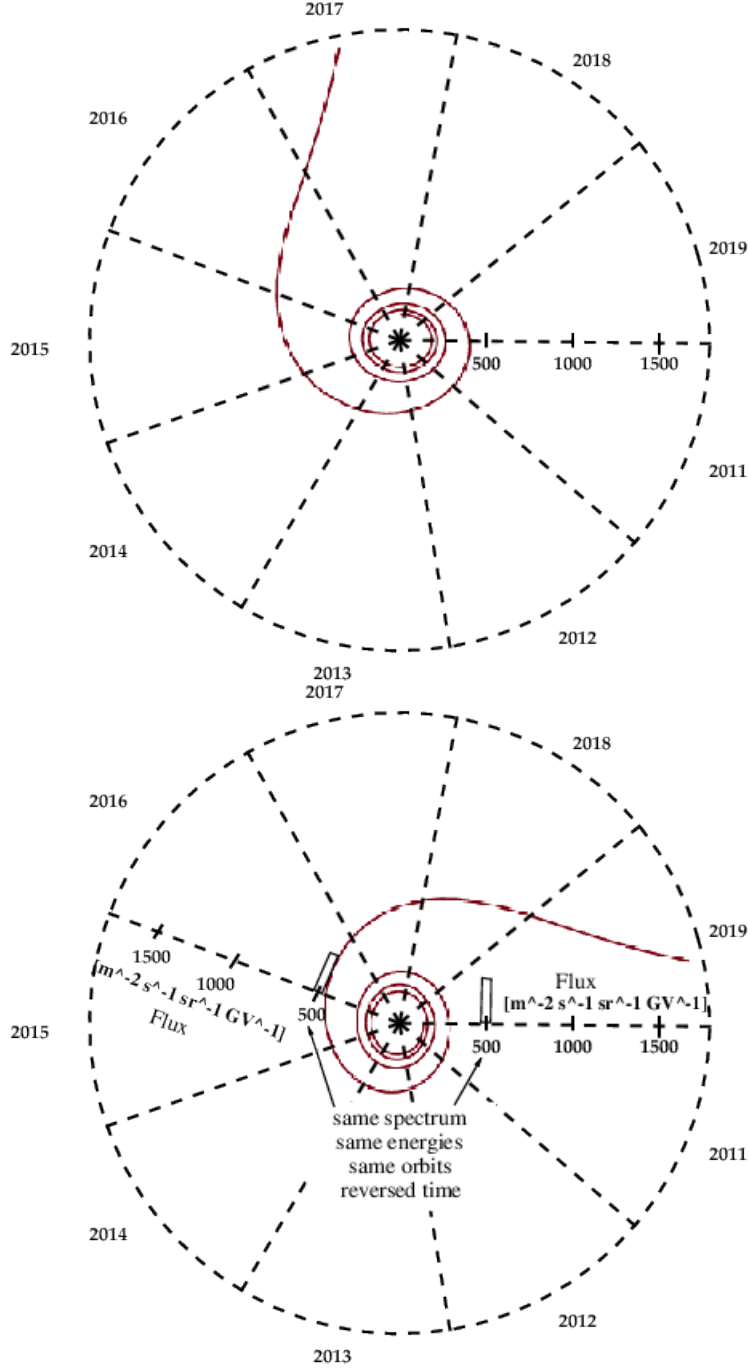


Figure 15: **Top:** Calculated (pseudo)-orbit for a proton departing from AMS for the sun in mid 2017. **Bottom:** Calculated (pseudo)-orbit (assuming negligible magnetic field in both cases) for proton departing from the sun for AMS in mid 2019.

The spiral orbits are calculated analytically lituus-modified spirals, (introduced by Galileo, who died the same year that Newton was born) as the orbit of a mass falling onto a massive sphere, which we, unlike Galileo's assumed Earth, are taking to be the sun. The lituus shape modification allows adjustment of the direction of the asymptotic straight line passing through the ISS. It is possible, but frustrating, to count the number of revolutions, even in this, almost the simplest possible case. In any case the orbit need not be permanently captured; out-of-plane velocity components cannot be seen in this projection.

As with all orbits observed by the ISS, these orbits do not represent single particles. They represent binned and sorted reconstructions with inferred energies having reliably small statistical sampling error, but no way of distinguishing the direction of travel.

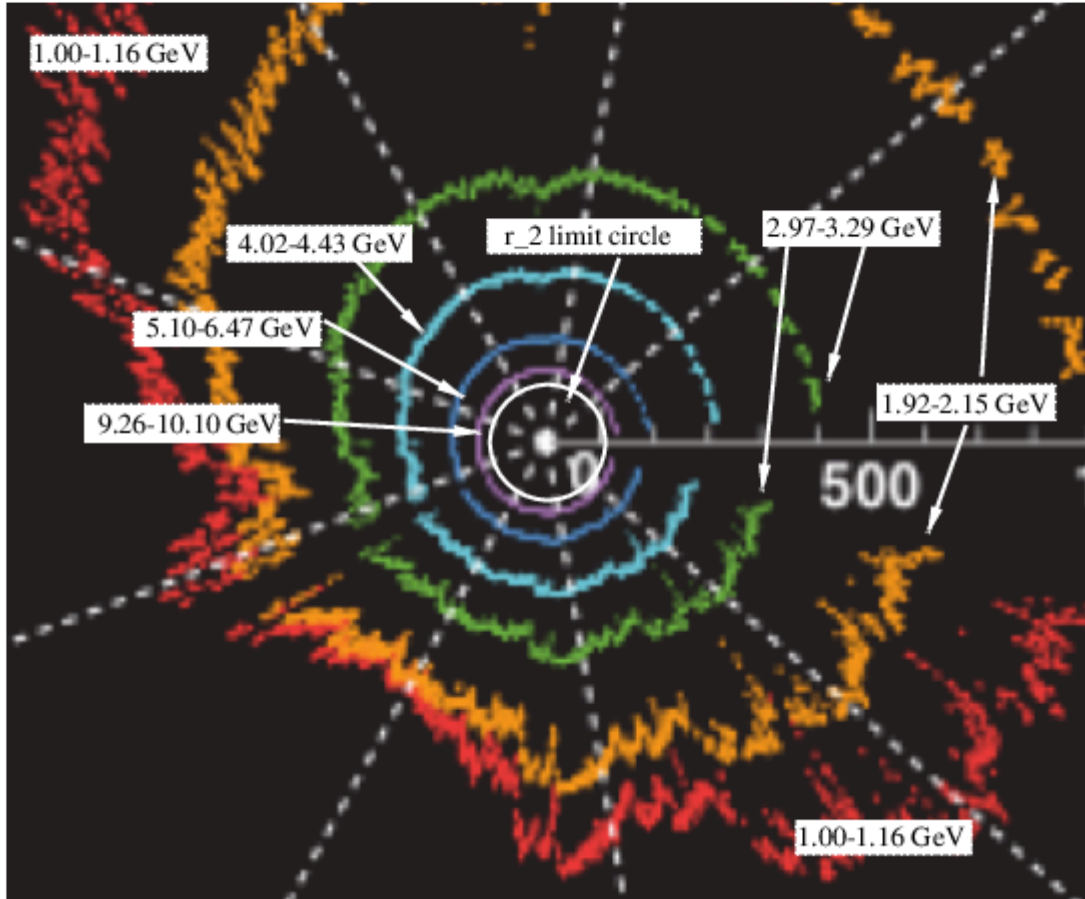


Figure 16: Central portion of a figure from M. Aguilar, et al. [16]. showing the daily AMS proton fluxes for six typical rigidity bins from 1.0 to 10.10 GV measured from May 20, 2011 to October 29, 2019 which includes a major portion of solar cycle 24 (from December 2008 to December 2019). The AMS data cover the ascending phase, the maximum, and descending phase to the minimum of solar cycle 24.)

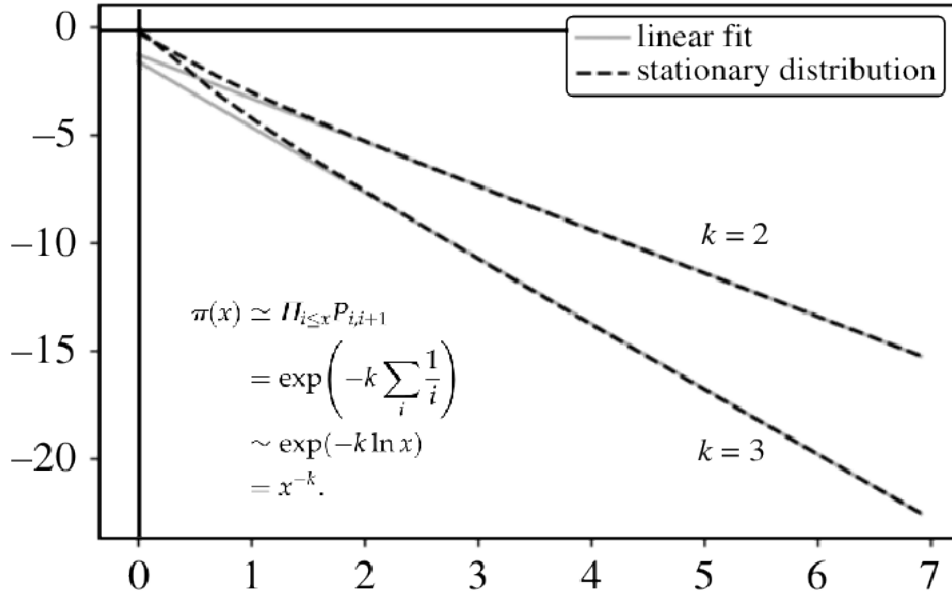


Figure 17: Copied from Roman and Bartoletti [10] “A master equation for power laws”, this figure shows how an iterated Markov stochastic distribution with one parameter (in this case, $k=2$) can match a different Markov process with parameter ($k=3$). Applied to our case, periodic perturbation of one Markov process, over a large (but finite) range of abscissas, can be accurately matched to a different Markov process. As explained in the text, this could account for the kinks in Figure (11), while continuing to treat the overall process as Markovian. In this way one can understand (as mathematics) why there need to be kinks at low energy but (as physics) there has to be an explanation for all fits having exactly the same slopes in the high energy limit, as shown in Figure (11).

APPENDICIES

A Relativistic energy and momentum

The relativistic principle of least action

It is straightforward to generalize the principle of least action in such a way as to satisfy the requirements of relativity while at the same time leaving non-relativistic relationships (i.e. Newton's Law) valid when speeds are small compared to c . Owing to the homogeneity of both space and time, the relativistically generalized action S cannot depend on the particle's coordinate 4-vector x^i . Furthermore it must be a relativistic scalar since otherwise it would have directional properties, forbidden by the isotropy of space.

The action of a free particle (i.e., one subject to no force) is

$$S = (-mc) \int_{t_0}^t ds = (-mc^2) \int_{t_0}^t \sqrt{1 - \frac{v^2}{c^2}} dt, \quad (45)$$

where the invariant interval ds is the proper time multiplied by c . As always, the dimensions of S are momentum $\times$ distance or, equivalently, as energy $\times$ time. Though the first expression for S is manifestly invariant, the second depends on values of $v \equiv |\dot{\mathbf{x}}|$ and t in the particular frame of reference in which Hamilton's principle is to be applied. *A priori* the multiplicative factor could be any constant, but it will be seen shortly why the factor has to be $(-mc^2)$. The negative sign is significant. It corresponds to the seemingly paradoxical result that the free particle path from position P_1 to position P_2 *maximizes* the proper time taken. Comparing with the standard definition of the action in terms of the Lagrangian, it can be seen that the free particle Lagrangian is

$$L(\mathbf{x}, \dot{\mathbf{x}}) = -mc^2 \sqrt{1 - \frac{|\dot{\mathbf{x}}|^2}{c^2}}. \quad (46)$$

As always, the Lagrangian has the dimensions of an energy.

In Lagrangian mechanics, once the Lagrangian is specified, the equations of motion follow just by "turning the crank". Slavishly following the Lagrangian prescriptions, the momentum $\mathbf{p}$ is *defined* by

$$\mathbf{p} = \frac{\partial L}{\partial \dot{\mathbf{x}}} = \frac{m\mathbf{v}}{\sqrt{1 - v^2/c^2}}. \quad (47)$$

For v small compared to c , this gives the non-relativistic result $\mathbf{p} \simeq m\mathbf{v}$. This is the relation that fixed the constant factor in the initial definition of the Lagrangian. From this equation one obtains the Hamiltonian H and hence the energy $\mathcal{E}$ by

$$H = \mathbf{p} \cdot \mathbf{v} - L = \frac{mc^2}{\sqrt{1 - v^2/c^2}}. \quad (48)$$

For v small compared to c , and the numerical value of H symbolized by $\mathcal{E}$, this gives

$$\mathcal{E} \simeq \mathcal{E}_0 + \frac{1}{2}mv^2, \quad (49)$$

which is the classical result for the kinetic energy, except for the additive constant $\mathcal{E}_0 = mc^2$, known as the rest energy. An additive constant like this has no effect in the Lagrangian description. From Eqs. (47) and (48) come the important identities

$$\mathcal{E}^2 = \mathbf{p}^2 c^2 + m^2 c^4, \quad \mathbf{p} = \frac{\mathcal{E}\mathbf{v}}{c^2}. \quad (50)$$

For massless particles like photons these reduce to $v = c$ and

$$p = \frac{\mathcal{E}}{c}. \quad (51)$$

This formula also becomes progressively more valid for a massive particle as its total energy becomes progressively large compared to its rest energy. As stated previously, m is the “rest mass”, a constant quantity, and there is no question of “mass increasing with velocity” as occurs in some descriptions of relativity, such as the famous “ $\mathcal{E} = mc^2$ ”, which is incorrect in modern formulations.

Remembering to express it in terms of $\mathbf{p}$, the relativistic Hamiltonian is given by

$$H(\mathbf{p}) = \sqrt{p^2 c^2 + m^2 c^4}. \quad (52)$$

4-vector notation

It can be seen that $\mathbf{p}$, as given by Eq. (47), and $\mathcal{E}$, as given by Eq. (48), are closely related to the 4-velocity u^i . We define a momentum 4-vector p^i by

$$p^i = m u^i = \frac{m}{\sqrt{1 - v^2/c^2}} \begin{pmatrix} c \\ \mathbf{v} \end{pmatrix} = \begin{pmatrix} \mathcal{E}/c \\ \mathbf{p} \end{pmatrix}. \quad (53)$$

We expect that $p^i p_i$, the scalar product of p^i with itself should, like all scalar products, be invariant. The first of Eqs. (50) shows this to be true;

$$p^i p_i = \mathcal{E}^2/c^2 - p^2 = m^2 c^2. \quad (54)$$

Belonging to the same 4-vector, the components of $\mathbf{p}$ and $\mathcal{E}/c$ in different coordinate frames are related according to the Lorentz transformation.

Forced motion

If the 4-velocity is to change, it has to be because force is applied to the particle. It is natural to define the 4-force G^i by the relation

$$G^i = \frac{dp^i}{ds/c} = \gamma \left(\frac{d\mathcal{E}/c}{dt}, \frac{d\mathbf{p}}{dt} \right)^T = \left(\frac{\mathbf{F} \cdot \mathbf{v}/c}{\sqrt{1 - v^2/c^2}}, \frac{\mathbf{F}}{\sqrt{1 - v^2/c^2}} \right)^T \quad (55)$$

where

$$\mathbf{F} = \frac{d\mathbf{p}}{dt} \quad (56)$$

is the classically defined force. Since this formula is valid both relativistically and non-relativistically it is least accident-prone 3D-form of Newton’s law. The energy/time component G^0 is related to the rate of work done on the particle by the external force. Note that this component vanishes in the case that $\mathbf{F} \cdot \mathbf{v} = 0$, as is true, for example, for a charged particle in a purely magnetic field.

B Hamilton-Jacobi wave-like particle mechanics

For simplicity this appendix discusses the simplest non-trivial application of Hamilton Jacobi wave-like mechanics, namely time independent, single particle mechanics.¹⁴ In this case a so-called “complete integral” of Hamilton’s partial differential equation exists. It is conventionally referred to as the “principal integral” or the “action function” $S(q_i, \alpha_i, t)$, in terms of which Hamilton’s first order differential equations can be obtained from the H-J partial differential equation.

$$\frac{\partial S}{\partial t} + H(q_i, \frac{\partial S}{\partial q_i}; t) = 0. \quad i = 1..3, \quad (57)$$

¹⁴For the present paper the single particle restriction will be strictly respected, but the time independence will need to be violated. Time dependence will be incorporated by exploiting the H-J separation of variables formalism; just like the Schrödinger, quantum mechanical treatment of the hydrogen atom. This will account for an occasional, once per cycle, brief energy acceleration. See Figure (8). This can account for an (adiabatically changing) periodic external accelerating force, while holding the other orbit elements constant.

Hamilton's differential equations can then be expressed in the form

$$\dot{p}_i = \frac{\partial S}{\partial q_i}, \quad \dot{\alpha}_i = -\frac{\partial S}{\partial \alpha_i}, \quad i = 1..3, \quad (58)$$

where α_1 , α_2 , and α_3 are (adjustable) constant parameters.

Using polar coordinates, the Lagrangian for a fundamental particle of variable mass $m_\beta \equiv m(\beta) \equiv m\gamma$, where $\gamma \equiv 1/\sqrt{1-\beta^2}$, and where, to begin with, the velocity $v = \beta c$ is assumed to be constant but, for example, highly relativistic, say $\beta = 0.98$ velocity, moving in three dimensions in the field of an inverse square law central gravitational force, and/or a transverse magnetic force, and/or a longitudinal electric force.

$$L = \frac{1}{2}m\gamma(\dot{r}^2 + r^2\dot{\theta}^2 + r^2\sin^2\theta\dot{\phi}^2) - \frac{K}{r}, \quad \text{where } K = \frac{Ze^2}{4\pi\epsilon_0 r}. \quad (59)$$

The canonical momenta are

$$p_r = m\gamma\dot{r}, \quad p_\theta = m\gamma\dot{\theta}, \quad p_\phi = m\gamma\sin^2\theta\dot{\phi}, \quad (60)$$

and the Hamiltonian is

$$H = \frac{p_r^2}{2m\gamma} + \frac{p_\theta^2}{2m\gamma^2} + \frac{p_\phi^2}{2m\gamma^2\sin^2\theta} + \frac{K}{r}. \quad (61)$$

For nuclear particles K and Z are both positive. Looking for a solution by separation of variables, the time-independent H-J equation is¹⁵

$$\mathcal{E} = \frac{1}{2m\gamma} \left(\left(\frac{dS^{(r)}}{dr} \right)^2 + \frac{1}{r^2} \left(\frac{dS^{(\theta)}}{d\theta} \right)^2 + \frac{1}{r^2\sin^2\theta} \left(\frac{dS^{(\phi)}}{d\phi} \right)^2 \right) + \frac{K}{r}. \quad (62)$$

Since ϕ does not appear explicitly we can separate it immediately in the same way t has already been separated;

$$S = -\mathcal{E}t + \alpha_3\phi + S^{(\theta)}(\theta) + S^{(r)}(r). \quad (63)$$

Here $\mathcal{E}$ is the first “new momentum” of Jacobi and α_3 is the second; it is interpretable as the value of a conserved azimuthal angular momentum around the z axis because

$$p_\phi = \frac{\partial S}{\partial \phi} = \alpha_3 \quad (64)$$

is constant. Substituting this into Eq. (62) and multiplying by $2m\gamma r^2$ yields

$$2m\gamma E r^2 - 2mK\gamma - r^2 \left(\frac{dS^{(r)}}{dr} \right)^2 = \left(\frac{dS^{(\theta)}}{d\theta} \right)^2 + \frac{1}{\sin^2\theta} \left(\frac{dS^{(\phi)}}{d\phi} \right)^2 = \alpha_2^2, \quad (65)$$

where the equality of a pure function of r to a pure function of θ implies that both are constant; this has permitted a third Jacobi parameter α_2 to be introduced. The physical meaning of α_2 can be inferred by expanding M^2 , the square of the total angular momentum;

$$M = \sqrt{(m\gamma r^2\dot{\theta})^2 + (m\gamma r^2\sin\theta\dot{\phi})^2} = \sqrt{p_\theta^2 + \frac{p_\phi^2}{\sin^2\theta}} = \alpha_2. \quad (66)$$

From the interpretation of α_3 as the z component of α_2 it follows that

$$\alpha_3 = \alpha_2 \cos i. \quad (67)$$

¹⁵As elsewhere in this paper, the conventional symbol E , for “energy” has been replaced by $\mathcal{E}$.

Determination of the other terms in S has been “reduced to quadratures” since Eqs. (65), gives expressions for $dS^{(\theta)}/d\theta$ and $dS^{(r)}/dr$ that can be re-arranged to yield $S^{(\theta)}(\theta)$ and $S^{(r)}(r)$ as indefinite integrals;

$$\begin{aligned} S_2 &= - \int^\theta \sqrt{\alpha_2^2 - \frac{\alpha_3^2}{\sin^2 \theta'}} d\theta', \\ S_3 &= \int^r \sqrt{2m\gamma E - \frac{2m\gamma K}{r'} - \frac{\alpha_2^2}{r'^2}} dr'. \end{aligned} \quad (68)$$

Instead of using E as the first Jacobi “new momentum” it is conventional to use a function of E , namely

$$\alpha_1 = \sqrt{\frac{-K^2 m \gamma}{2E}}, \quad E = \frac{-K^2 m \gamma}{2\alpha_1^2}. \quad (69)$$

Like α_2 and α_3 , α_1 has dimensions of angular momentum. The semi-major axis a and the orbit eccentricity ϵ are given by

$$a = -\frac{\alpha_1^2}{Km\gamma}, \quad 1 - \epsilon^2 = \left(\frac{\alpha_2}{\alpha_1}\right)^2, \quad (70)$$

with inverse relations

$$\alpha_1^2 = -Km\gamma a, \quad \alpha_2^2 = -(1 - \epsilon^2)Km\gamma a. \quad (71)$$

Combining results, the “action integral” of the H-J equation is

$$S = \frac{m\gamma K^2}{2\alpha_1^2} t + \alpha_3 \phi - \int_{\pi/2}^\theta \sqrt{\alpha_2^2 - \frac{\alpha_3^2}{\sin^2 \theta'}} d\theta' + \int_{a(1-\epsilon^2)}^r \sqrt{-\frac{(m\gamma)^2 K^2}{\alpha_1^2} - \frac{2m\gamma K}{r'} - \frac{\alpha_2^2}{r'^2}} dr'. \quad (72)$$

The lower limits and some signs have been chosen arbitrarily so that the Jacobi “new momenta” β_1 , β_2 and β_3 will have conventional meanings. Refer to reference [18] for further details.

The purpose of this appendix has been to reformulate classical relativistic particle mechanics as wave mechanics, from which single particle orbits, treated as “rays”, are unambiguous and calculable. Mathematically, the classical orbits are stationary time “rays” normal to H-J “wave-fronts”.

This formalism was established by Jacobi almost a century before Einstein’s special relativity and longer than that before Heisenberg’s and Schrödinger’s non-relativistic quantum mechanics. It is more or less equivalent to the Bohr-Sommerfeld model of the hydrogen atom.

During the years from 1900 to 1920, Jacobi’s wave theory, (developed a century earlier) as well as being consistent with special relativity, was surprisingly similar to elementary, non-relativistic quantum mechanics about to be developed—absent, of course, failure of commutation, particle creation, and Heisenberg uncertainty.

For the present cosmic ray investigation, the H-J theory seems to be especially well matched. In the context of this paper, the first invaluable feature of the separation of variables method of solution of the Hamilton-Jacobi solution is that only one “separable momentum”, the particle energy energy changes while a particle is circulating about the sun. During this time interval the other three “momenta” remain constant. The second invaluable feature is that, while out of contact with the sun, the energy remains constant, while the other three momenta change, in accordance with H-J theory, plus stochastic dependence on stray magnetic fields.

As one result of the existence of these alternating states of motion, the radial dependence of the energy is highly predictable, with potential energy depending only on radial position, r . Meanwhile, the transverse deviations from centripetal or centrifugal motion are caused by magnetic fields that, though weak, are essentially unknown. Theoretical description of these motions needs to be described by adiabatic variation of the other three separation “constants”, as constrained by the known energy variation.

Since the magnetic fields are, at best, known only in some statistical sense, the transverse deviations are necessarily knowable only in a stochastic sense. The basically unpredictable wandering caused by magnetic fields needs to be modeled stochastically.

While in contact with the sun, the motion is governed by the magneto-hydrodynamic properties of the solar plasma. This, too, is also stochastic, but the energy gain or loss is well represented by the total energies of individual particles or coherent bunches of particles.

C Kulsrud et al. theory of cosmic ray, stellar plasma interaction

Never having believed that cosmic rays were galactic, let alone extra-galactic in origin, I have, until recently, ignored papers assuming extra-solar sources; which is to say ignoring the entire plasma theory of the sources of cosmic rays. I have only belatedly come to realize that the physics of the source of cosmic rays is more or less the same, irrespective of their source location, be it solar or extra-galactic. Understanding only accelerators, and certainly not magneto-hydrodynamic, this attitude has not been particularly appropriate.

Having overcome this impediment, I have found that reference[19] and Kulsrud's book[20], "Plasma Physics for Astrophysics", in particular Chapter 12, have provided answers to questions that I have been unable even to formulate effectively myself. Because of its amazing relevance, and to avoid symbol clashes, this appendix exhibits and comments on paragraphs copied along with figures from these sources. What makes this promising is the surprising way in which plasma physics can "imitate" laboratory equipment that itself took half a century to engineer.

This comment applies especially to particle injection into terrestrial accelerators which, though adequately understood for most practical purposes, continues to be a serious research area to this day. For both laboratory and celestial accelerators, beam extraction is easier to understand than injection, and hence less controversial.

Initially following Kulsrud, This appendix attempts to compare laboratory and celestial accelerators, at least superficially, and as briefly as possible.

The second paragraph of Chapter 12 of Kulsrud's book begins as follows:

"Let the number of cosmic rays in the energy range $d\epsilon$ be $N(\epsilon)d\epsilon$. Their energy spectrum $N(\epsilon)$ is very nearly a perfect power law, with exponent -2.7 from 1 to 10^6 GeV

$$N(\epsilon) \sim \epsilon^{-2.7}.$$

Above 10^6 GeV it remains a power law, but the exponent changes to 3.1 and then back again to 2.7 above 10^6 GeV. It is believed these very high energy cosmic rays are extra-galactic and fill the universe uniformly. Their origin is believed to be in galactic sources."

It should go without my saying that I agree with the first half of this statement, which conforms closely with most of this paper, but categorically disagree with the latter half.

Kulsrud's following several paragraphs explain why inter-galactic cosmic ray scattering can be neglected without any serious concern. This means that the cosmic ray sources must lie within galaxies. *As an aside, my approach goes further, by saying the sources are almost entirely within the solar system.* At the same time, The existence of galactic cosmic rays into the solar system seems to be required for "start up" of the solar cosmic ray factory. Kulsrud discusses cosmic abundance ratios and many other issues in detail.

Especially significant is Kulsrud's section 12.2, "Pitch-angle Scattering of Cosmic Rays by Alfvén Waves". This topic is described in greater detail in Felice and Kulsrud[19], from which Figure (18) is copied. The abstract to this paper begins with the lines

"We study the problem of cosmic-ray diffusion in the Galactic disk with particular attention paid to the problem of particle scattering through the pitch angle in momentum space by wave-particle mirror interaction (here v is the cosmic-ray velocity parallel to the average Galactic magnetic field)."

The left figures is just as applicable to our sun as it would be to any star in any galaxy. It is matched by Figure 18, for which the top figure shows two cells of an Alvarez proton linac. Each cell is resonant at the same frequency but is tuned in phase to produce a slow (i.e. less than the speed of light) traveling electromagnetic wave, moving at exactly the average speed of “captured” bunches of protons. Each captured proton follows an elliptical paths that appears to be closed, like the two shown, in the frame of reference of the captured protons.

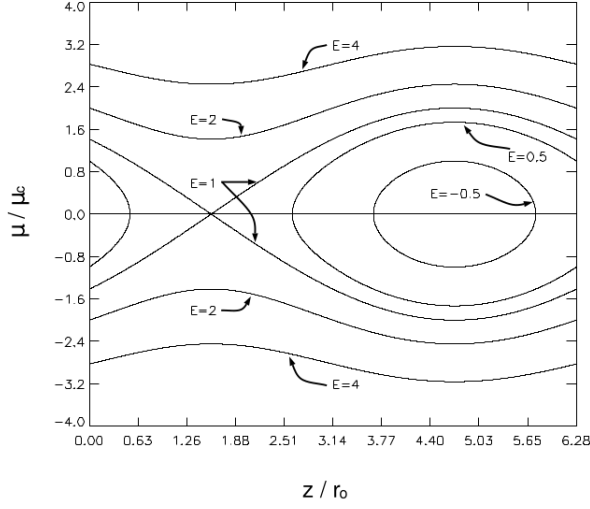


FIG. 1.—Contour plots of the particle energy. Note the presence of a separatrix at $E = 1$ and the trap region for $E < 1$.

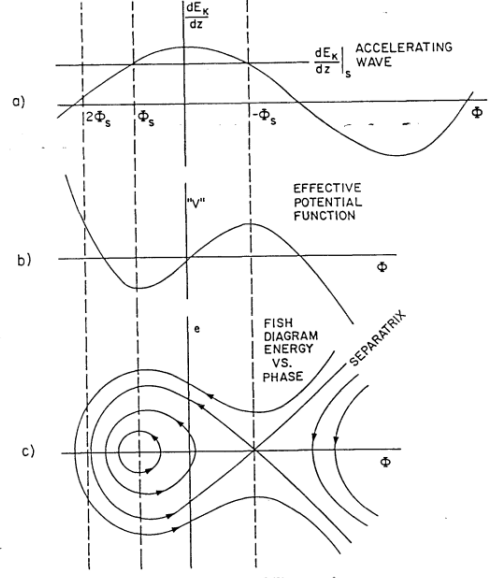


Fig. 6. Phase stability graphs.

Figure 18: **Left:** Copied from Felice and Kulsrud [19], along with its original caption, longitudinal phase space in plasma atmosphere of a star, and **Right:** copied from Loew and Talman [21] longitudinal phase space in a linear proton “Alvarez” accelerator. The similarity of these two figures suggests that the physics of the acceleration of cosmic rays near a star resembles the physics of acceleration of electrons in a linear accelerator. In each case there can be charges trapped in moving “stable buckets”. A figure on page 347 of reference [20] resembles the upper two figures on the fight, and is followed by appropriate explanation of the plasma physics.

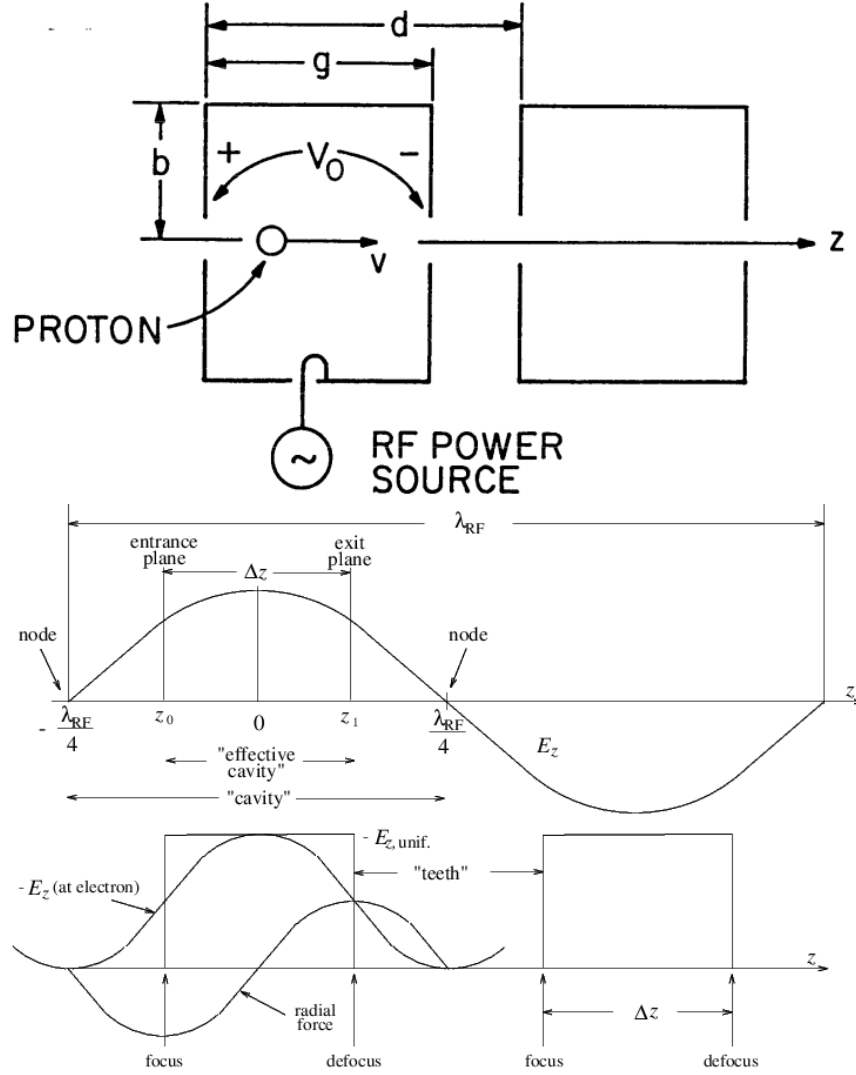


Figure 19: Copied from Loew and Talman,[21] the **upper** figure shows one of the many RF-cavity accelerating cells of an Alvarez linac needed to produce a “slow” (meaning slower than the speed of light) electromagnetic traveling wave. Individual cells, though seemingly independent, are phased to produce the desired wave speed, for example such that a moving proton is always “on crest”. The **lower** figure resembles Figure 12.1 in the Kudsrud book[20] which shows plasma fields capable of accelerating cosmic rays in galactic magneto-hydrodynamic Alfven waves. This figure, applicable to cells in an Alvarez linac, models a proton’s instantaneous position and replacement of actual accelerating field by a train of uniform field “teeth” of thickness $\Delta z \simeq \lambda_{RF}/4$. Any radial force is modeled by delta functions at entrances and exits.

D Accelerator physics units

With just a single exception, (with unfortunately confusing consequences) “accelerator physics units” and MKSA units are interchangeable, much in the same way that one meter is 100 centimeters. The exception is that energies are measured in GeV, rather than eV. What makes this change popular for high energy accelerator and particle physicists is that, within $\pm 15\%$, the proton mass (expressed as rest energy), or by nuclear A-value, or by atomic mass units, is roughly equal to 1. Electric fields are measured in GeV/m, rather than V/m, which is not inconvenient. Another easily-remembered practical result is the Boltzmann conversion from Kelvin temperature to energy; an energy of 1 MeV corresponds to a temperature of 1.16×10^{10} K.

However, the units for many other physical quantities become somewhat obscure; especially the magnetic field, because the Lorentz magnetic force is proportional both to magnetic field and to particle velocity. Checking that electric and magnetic fields are close to what is expected, it should be possible to interpret all physical parameters correctly.

Incidentally, these accelerator units are much like “natural units” in that the speed of light is equal to 1. In this paper SI formulas involving the speed of light retain the factor “c”, but its numerical value is 1.0. The fact that the Planck (modified) constant $\hbar$, is not set equal to 1 in accelerator units complicates this practice. For analyzing weak interaction processes, natural units are greatly to be favored.

In the present paper, accelerators with superimposed electric and magnetic fields are first described, with bending fractions η_E and η_M which sum to one. This is followed immediately by celestial accelerators having predominantly gravitational bending with similar bending fractions η_G^* and η_M^* , which are proportional to the bending strengths, but do not sum to 1.

For E&M bending, to avoid the direct evaluation of magnetic fields in Tesla units, electric fields are given in GV/m, while magnetic bending fields are inferred from η_{M1} the magnetic bending fraction for beam 1. For celestial G&M bending a similar convention has needed to be devised.

References

- [1] R. Talman, *Solar cosmic ray generation Newtonian gravity, missing mas, dark energy laboratory-based astrophysics, and all that*, arXiv:2508.19296v1 [physics.acc-ph] 25 Aug 2025
- [2] M.J. Owens and R.J. Forsythe, *The Heliospheric magnetic field*, doi:10.12942/lrsp-2013-5
- [3] J.J. Beatty, J. Matthews, and S.P. Wakely, *Cosmic rays*, Chapter 29 of Particle Data Group Handbook, 2019
- [4] C.R. Cramer, *The Sun’s Alfvén Surface*, arXiv.2310.05887v1 [astro-ph.SR] 9 Oct 2023
- [5] S. R. Crammer et al., *The sun’s Alfvén surface: recent insights, and prospects for the polarimeter to unify the corona and heliosphere*, arXiv:2310.05887v1 [astro-ph.SR] 9 Oct 2023
- [6] M. Aguilar, et al. Antiprotons and Elementary Particles over a Solar Cycle: Results from the Alpha Magnetic Spectrometer, PRL 134, 051002, 2025
- [7] E. G. Blackman, *The magnetic field of Jupiter*, www.pas.rochester.edu,
- [8] S.A. Koldobsky, *Time Lag Between Cosmic-Ray and Solar Variability: Sunspot Numbers and Open Solar Magnetic Flux*, <https://doi.org/10.1007/s11207-022-01970-1>, 2021
- [9] Richard Talman, *Geometric Mechanics*, 2nd edition, Wiley VCH, Chapter 16, 2007
- [10] S. Roman and F. Bertolotti, *A master equation for power laws*, <https://doi.org/10.1098/rsos.220531>, 2022

- [11] Richard Talman, *Talman-Accelerator X-ray-Sources*, Wiley VCH, Chapter 3, 2006
- [12] Richard Talman, *Geometric Mechanics*, 2nd edition, Wiley VCH, Chapter 14, 2007
- [13] K. H. Schatten, J. M. Wilcox, and N. F. Ness, *A model of interplanetary and coronal magnetic fields*, Solar Physics 6 (1.969) 442-455 1968
- [14] R. Talman, *Novel Relativistic Effect Important in Accelerators*, PRL **56**, 14, p.1429, 1986
- [15] P. Lipari, and S. Vernetto, *The shape of the cosmic ray proton spectrum*, Astroparticle Physics, 120, 102441, 2020
- [16] M. Aguilar, et al. *Periodicity's in the Daily Proton Fluxes, Alpha Magnetic Spectrometer (AMS)*, PRL 127, 271102, 2021
- [17] G. Munoz and I. Pavic, *A Hamilton-like vector for the special-relativistic Coulomb problem*, Eur. J. Phys. 27 (2006) 1007–1018, doi:10.1088/0143-0807/27/5/001
- [18] Richard Talman, *Geometric Mechanics*, 2nd edition, Wiley VCH, Chapter 8, 2007
- [19] G. M. Felice and K.M Kilsrud, *Cosmic-ray pitch-angle scattering through 90 degrees*, Astrophysical Journal 553, 198
- [20] Russell M. Kulsrud, *Plasma Physics for Astrophysics*, especially Chapter 12, Princeton University Press, 2005
- [21] G. A. Loew and R. Talman, *Elementary Principles of Linear Accelerators*, SLAC-PUB-3221, 1983